Rank-six fusion rings: categorification criteria and models through multiplicity eight

23 September 2026

Abstract

We study complex categorifiability of the 16612 rank-six fusion rings of multiplicity at most eight in the *Fusion Ring Census*. Combining arithmetic and subring obstructions with a unified central-induction framework, we obtain 16511 general exclusions, 54 categorifiable rings, and 47 unresolved cases. Categorifiability is completely classified through multiplicity two: 43 of 193 rings admit spherical models, including 42 with unitary models. Our necessary criteria and model identifications are applied automatically, with exact certificates and a stepwise exclusion table. For the first unresolved ring, at multiplicity three, we prove and test a fourth-order product-Gram condition for pseudo-unitary centers. It survives this test but admits no braided spherical categorification, by the Dong–Schopieray catalogue; its general categorifiability remains open.

Contents

1 Classification results and scope **1**

2 Census input and tensor conventions **4**

3 Arithmetic and categorical reductions **5**

4 One central-induction certificate framework **10**

5 Module categories and Morita constraints **24**

6 Categorical models and exact tensor reconstruction **30**

7 Certificates, research coverage, and the remaining frontier **50**

1 Classification results and scope

A fusion ring is a finite-rank based ring with nonnegative integral structure constants, a distinguished unit, and duality satisfying Frobenius reciprocity. We work with full ordered multiplication tensors; in particular, commutativity is not part of the definition. The multiplicity is the maximum of all structure constants, including the unit coefficients. A categorification means a fusion category over $\mathbb{C}$, without an assumed pivotal, spherical, or unitary structure. We classify existence for a ring, not equivalence classes of categories with that ring.

Vercleyen–Slingerland constructed a substantial low-rank database [42]. Its rank-six portion is complete through multiplicity four. The multiplicity-one categorification problem was settled by Liu–Palcoux–Ren [27]. The rank-four and rank-five projects [12, 11] develop arithmetic, sphericalization, and central-induction certificates. At rank six, however, neither the ring nor

a signed quotient of its sphericalization can simply be assumed commutative. This is the principal structural change addressed here.

All positively identified rings and their models appear in Table 3; its structure column is part of the classification statement, not merely a description of a chosen presentation. In particular, Schopieray's category [41, Example 1] realizes $R^{(6)}_{2,124}$ spherically, whereas [41, Theorem 1] excludes *every* pseudo-unitary categorification of that same ring. Thus general or spherical existence cannot be identified with unitary existence, even in rank six and multiplicity two. The electronic ledger records general, pivotal, spherical, pseudo-unitary and unitary existence separately for every tensor. The source identifications are detailed in Section 6.4. The paper is organized by mathematical operation rather than by individual exceptions. Section 2 fixes the source census, its scope, and the tensor conventions used by every automated test. Section 3 reduces arbitrary categorifications to the appropriate arithmetic and spherical cases. Section 4 presents one central-induction framework, with precisely scoped refinements rather than separate ad hoc tests. Section 5 supplies module and Morita constraints. Section 6 treats models by construction, including central-boson quotients, equivariantizations, and relative centers. Section 7 specifies exhaustive certificate checking and the remaining frontier.

In particular, the positive catalogue includes the extended Haagerup category $\mathcal{EH}_1$ [4, 6] and Izumi's $C_5 + 5$ near-group category [23], not just the constructions derived from them. The mixed fermionic quotient of Proposition 6.17, the sign-swap equivariantization of Theorem 6.18, and the relative-center construction of Corollary 6.20 give three additional full-tensor identifications. None is inferred merely from a dimension match or from survival of the necessary criteria.

Theorem 1.1 (Source census [37])**.** *The published* Fusion Ring Census*, at the revision fixed in Section 2, contains exactly* 16612 *based-isomorphism classes of rank-six fusion rings of multiplicity at most eight. The numbers at exact multiplicities* $1, \ldots, 8$ *are*

$$39, \quad 154, \quad 384, \quad 872, \quad 1582, \quad 2768, \quad 3814, \quad 6999.$$

All three possible duality types and all noncommutative cases are included. No categorifiability filter is applied. Completeness of this counting input is the result of the cited repository; the present paper addresses categorification of its tensors.

Theorem 1.2 (Classification through multiplicity two and certified extension)**.** *In the census of Theorem 1.1, the accompanying exact certificates exclude* 16511 *rings from general complex categorification. There are* 54 *positively identified rings:* 53 *have unitary categorifications, whereas* $R^{(6)}_{2,124}$ *has a spherical categorification but no pseudo-unitary categorification. These positive sets and the general exclusions are pairwise disjoint. The remaining* 47 *rings are not decided for general existence by the present criteria and bounded searches. Structure-specific pseudo-unitary exclusions are recorded separately and do not alter that general status. Their distribution is given in Table 1.*

Among the 193 *rings of multiplicity at most two, precisely* 43 *admit general, pivotal, or spherical categorifications. Exactly* 42 *of these admit pseudo-unitary or unitary categorifications; the sole exception is* $R^{(6)}_{2,124}$*. At exact multiplicity two the* 154 *rings split into* 132 *general exclusions,* 21 *unitary positives and that one spherical non-pseudo-unitarily-categorifiable ring. There are no residuals at multiplicity two. The higher-multiplicity results in Table 1 are partial; their residuals are listed in the complete electronic ledger.*

The low-multiplicity residuals. Every ring at exact multiplicity 2 is decided. At exact multiplicity 3 the unresolved records are

$$R^{(6)}_{3,215}.$$

Exact multiplicity	Rings	Excluded	Unitary	S, not PU	Undecided
1	39	18	21	0	0
2	154	132	21	1	0
3	384	380	3	0	1
4	872	855	2	0	15
5	1582	1577	1	0	4
6	2768	2762	3	0	3
7	3814	3807	2	0	5
8	6999	6980	0	0	19
Total	16612	16511	53	1	47

Table 1: Complete counting census; categorification is complete through multiplicity two and partial above. The column S, not PU denotes a spherical realization with no pseudo-unitary categorification. All excluded rings are ruled out generally.

The phrase "undecided" is deliberately local to this computation. A search timeout is not an obstruction, and failure to produce a separating functional is not a construction. Nor do we claim that every residual is open throughout the literature. Section 7.2 retains exact feasible points for the now positively identified cyclic-four ring as regression controls. Such points were not used to infer existence; the conformal-embedding construction supplies it.

For reproducible identifiers, retain the unit as the zeroth basis element, use the exported representatives in the generator's fixed coordinate conventions, and sort their full tensors in row-major lexicographic order at each exact multiplicity. This order is a convention on the exported representatives, not a claim that each is the lexicographic minimum under all 120 relabellings. Distinctness up to based isomorphism is supplied by the cited census. The established representatives and identifiers through multiplicity seven are preserved; the multiplicity-eight stratum is sorted by the same row-major convention. The jth tensor at multiplicity m is denoted $R^{(6)}_{m,j}$. The complete tensor, rather than this identifier, is the input to every obstruction. A separate crosswalk records matches to the ancillary database of [42]; its indices play no role in a proof.

All reductions are applied to full tensors. The cheap arithmetic filters are retained because they avoid unnecessary induction searches; the stronger blocks are used only with their stated hypotheses. The rank-four and rank-five results are cited rather than reproved. This is a unified implementation of sufficient exclusion certificates, not a claim that the criterion set is logically minimal.

Progress at the first unresolved multiplicity. The earliest general residual remains $R^{(6)}_{3,215}$. We do not skip it in favor of larger-multiplicity examples. Section 4.6 proves a joint low-product Gram condition, combining fourth-order modular moments with integral product restrictions. Its exact automated test shows that the formal rank-22 witness survives; this is a precise limit of the strengthened necessary system, not a new general exclusion. Section 3.5 derives from Dong–Schopieray a rank-six bound $\mathrm{FPdim} \leq 168$ for braided spherical categorifications. Since the tensor has $\mathrm{FPdim} = 136 + 32\sqrt{17} > 168$, it is excluded in that scope only. A full-census tensor comparison with the rank-six ancillary data of Ng–Rowell–Wen gives no new model. Each numerical conclusion is produced by the supplied scripts from tensors and exact source data, without manual case arithmetic.

What the exclusion table measures. The sequential table in Section 7.1 applies the available criteria in an explicit order and reports new removals and remaining tensors at each exact multiplicity. Simple structural and arithmetic tests precede finite induction and signed

searches. This is a reproducible compromise between cost and yield, not a proof of a universally optimal order. A theorem with a stronger conclusion can be more expensive to apply, and a successful earlier test makes later tests unnecessary for that tensor. Dimension-order and character-induction stages first make additional top-level removals at multiplicity eight, one and four respectively. This describes the executed sieve through eight; it does not claim that the same criteria decide larger multiplicities.

The earliest unresolved multiplicity is the first stratum in which the current automated procedure does not decide every ring. It marks where further work should begin; that work may be a new exclusion criterion, a sharper application of an existing criterion, or a positive model. A zero residual count in a later stratum means that the present methods suffice for that stratum. Even if an entire terminal interval through multiplicity eight has zero residuals, this would not imply that no new criterion is needed at multiplicity nine or higher. Conversely, a bounded search that stops without a certificate does not show that a new theorem is logically necessary.

The same reporting scheme applies at lower ranks: one fixes a complete counting range, runs the necessary criteria and model constructors, and locates the first nonzero residual stratum. Here the detailed automated ledger is rank six. The cited rank-four and rank-five categorification results are used only in their proved ranges; the larger counting ranges of the source repository do not by themselves extend those theorems.

2 Census input and tensor conventions

2.1 One authoritative counting source

All counting data are taken from the public repository *Fusion Ring Census* [37], pinned to revision

`b86eb6dc6eeeafe5fc5f6906d7b074e0322e9726`.

The source files are `results/census.json` and `results/rank6/tables.txt.gz`. They declare a complete, verified rank-six census through multiplicity eight. Its 16612 tensors include all 9613 classes through multiplicity seven and 6999 classes of exact multiplicity eight. The exact-multiplicity definition, inclusion of the unit in the rank, and inclusion of every duality type are unchanged.

The paper trusts this census as its counting input. It supplies neither another census generator nor a claim to have independently enumerated the same domain. The repository contains the enumeration algorithms, coverage arguments, completion records, and tensor checks supporting Theorem 1.1. A valid and duplicate-free tensor list alone would not establish exhaustive coverage; that claim is supplied by the source census, not by the import operation below.

2.2 Frozen data and reproducible identifiers

A copy of the source tensor export is retained with the classification code, so a classification check has no live-network dependency. Its compressed SHA-256 is

`0138c6228dbb06b0861ad38b8dfe67fcd1fc829add80d441c34c09a1afb78675`,

and its uncompressed SHA-256 is

`7898a833dfffd0f645436c1d326740db29ea6a29980de307c67227f0b67122b0`.

These hashes identify the input bytes; they are not substitutes for the source's completeness argument or for categorification certificates.

Every ring is represented by its full ordered tensor $N = (N_{ij}^k)_{0 \le i,j,k < 6}$, with $b_0 = \mathbf{1}$. The multiplicity is $\max_{i,j,k} N_{ij}^k$. Noncommutative inputs are retained without symmetrization. A based-ring isomorphism is a unit-fixing basis permutation preserving every structure constant. Consequently, comparison with a model uses the complete tensor, not only dimensions, a fusion graph, or a numerical fingerprint.

The identifier $R_{m,j}^{(6)}$ is local to this paper's frozen tensor registry. Representatives and identifiers through multiplicity seven are retained. The new exact-multiplicity-eight tensors are placed in row-major lexicographic order using the exported representatives. This order need not give the lexicographically smallest representative under all basis permutations. Import matching records the correspondence between the source representatives and the registry. Every mathematical criterion receives the tensor rather than the identifier, so relabelling cannot alter its conclusion.

2.3 Counting, exclusions and constructions

The source census answers a finite counting question about based rings. A categorification exclusion answers a different question about complex fusion categories. We therefore keep the census fixed while applying necessary criteria and matching models. The resulting ledger covers every imported tensor exactly once; accepted negative certificates, positive constructions and unresolved records are disjoint.

The exact prefix $39, 154, 384, 872$ agrees with the complete rank-six range of [42]. We use no external categorifiability flag as a proof. The classification code may reconstruct arithmetic, tensor or isomorphism data required by a criterion, but it does not repeat the census search. The counting proposal in `oeis/` derives directly from the same source counts and has no dependence on whether a categorification has been found.

3 Arithmetic and categorical reductions

3.1 Conventions and imported criteria

Write N_{ij}^k for the coefficient of b_k in $b_i b_j$ and i^* for duality. The matrices used in the software have entries $(N_i)_{jk} = N_{ij}^k$. Consequently they give the right-handed matrix convention

$$N_i N_j = \sum_k N_{ji}^k N_k.$$

This distinction is harmless for commuting matrices, but essential when checking noncommutative signed tables. Frobenius reciprocity gives $N_{i^*} = N_i^{\mathsf{T}}$. The positive dimension character is denoted $d_i = \mathrm{FPdim}(b_i)$ and $D = \sum_i d_i^2$.

The complexified based ring is a semisimple $*$-algebra. For an irreducible representation E of dimension r_E, let f_E be its formal codegree, in the convention of [35]. The regular Casimir matrix

$$C_R = \sum_i N_i N_{i^*}$$

has eigenvalue $r_E f_E$ with multiplicity r_E^2. It is only in the commutative case that all these multiplicities are one and its characteristic roots are themselves the formal codegrees.

We use the following established necessary conditions. Their precise applications are certified by polynomial, prime, dimension-field, and signed-table witnesses.

Proposition 3.1 (Established inputs)**.** *For a complex fusion categorification, the following hold.*

(i) Every formal codegree is a d-number [35]. If $p(t) = t^s + a_1 t^{s-1} + \cdots + a_s$ is its minimal polynomial, this is expressed by $a_s^i \mid a_i^s$ for $1 \le i \le s$.

(ii) For a commutative Grothendieck ring, character values are cyclotomic integers. The polynomial factorization and finite-field witnesses used here are those of [12, Section 2]; the commutativity hypothesis is checked before this test is used.

(iii) A fusion subring gives a fusion subcategory. Its Frobenius–Perron dimension divides that of the ambient category in the ring of algebraic integers. A subring of rank two with $x^2 = 1 + mx$, $m \ge 2$, is excluded by [34].

(iv) Weakly integral fusion categories are pseudo-unitary [14]. For the two-Frobenius–Perron-dimension reduction we use the Galois-conjugacy result of [39, Lemma 3.1]. It is a reduction to a pseudo-unitary Galois conjugate, not a claim that every originally given spherical structure is positive.

(v) The zero-spectrum obstruction of [27, Theorem 2.14] is a general categorification obstruction. Its finite hypothesis is tested directly on the full ordered tensor.

(vi) The spherical dimension, normalized norm, sign-object moment, adjoint-parity, and low-induction criteria are those of [12, Sections 2,3,6,7]. The finite joint relative-trace criterion is the named theorem of [11].

For completeness, the finite tensor hypothesis in input (v) is as follows. Put $\langle ab, cd\rangle = \sum_k N_{ab}^k N_{cd}^k$. There must be nine indices $a, b, c, u, v, w, x, y, z$ such that

$$N_{ua}^w N_{vu}^b N_{vw}^c N_{xz}^a N_{bx}^y N_{yz}^c > 0, \quad N_{ba}^c = 1, \quad \sum_k N_{ux}^k N_{v^*y}^k N_{wz^*}^k = 0,$$

and each of the two sets

$$\{\langle vu, ca^*\rangle, \langle bu^*, cw^*\rangle, \langle v^*b, wa^*\rangle\}, \qquad \{\langle bx, cz^*\rangle, \langle yx^*, ca^*\rangle, \langle b^*y, az^*\rangle\}$$

contains 1. Theorem 2.14 of [27] excludes any such tensor. The certificate stores the nine indices and the checker reevaluates all these expressions. No associator is assumed to be unitary or pivotal.

Input (v) is especially inexpensive because it needs just a small integer witness. We do not import the negative part of any integral classification table or quadratic-family classification as a list of forbidden labels.

Corollary 3.2 (Certified lower-rank descent)**.** *Suppose a rank-six fusion ring contains a based fusion subring of rank four and multiplicity at most* 128*, or of rank five and multiplicity at most* 20*. If that subring is not based-isomorphic to one of the positive rings in the respective main theorem of [12, 11], the rank-six ring is not categorifiable over $\mathbb{C}$.*

Proof. A hypothetical categorification would contain the fusion subcategory on the prescribed basis subset. Apply the cited complete lower-rank classification. The test enumerates unit-containing dual-closed subsets, checks product closure, and compares the restricted tensor with every positive model under every unit-fixing permutation. No negative labels are read. □

3.2 A certificate of a nonabelian character field

Theorem 3.3 (Noncommuting field automorphisms)**.** *Let R be a commutative fusion ring. Suppose an irreducible factor $f \in \mathbb{Q}[t]$ of the characteristic polynomial of some multiplication*

matrix admits polynomials $a, b \in \mathbb{Q}[t]$ such that

$$f(a(t)) \equiv f(b(t)) \equiv 0 \pmod{f}, \qquad a(b(t)) - b(a(t)) \not\equiv 0 \pmod{f}.$$

Then R has no complex fusion categorification.

Proof. In a categorification, the eigenvalues in question are values of one-dimensional characters of $R \otimes \mathbb{C}$ and are cyclotomic integers by Proposition 3.1(ii). Hence $K = \mathbb{Q}[t]/(f)$ embeds into an abelian cyclotomic extension of $\mathbb{Q}$. Every subfield of an abelian Galois extension is Galois with abelian Galois group. Substitution by a and b defines unital endomorphisms of the field K; they are injective, hence surjective by finite dimensionality. The last displayed condition says that these automorphisms do not commute, a contradiction. □

Factoring f over $\mathbb{Q}[t]/(f)$ is allowed in discovery, but is unnecessary in verification: irreducibility, divisibility, two substitutions and one nonzero remainder suffice. In the present census the test excludes $R^{(6)}_{2,119}$, with the degree-six factor

$$f(t) = t^6 - 3t^5 - 2t^4 + 9t^3 - 5t + 1.$$

The two substitution polynomials and their nonzero commutator are stored in the certificate and recomputed from this polynomial. This test is a stronger certificate generator for the existing cyclotomicity condition, not a new categorical hypothesis.

The same automated test excludes $R^{(6)}_{6,1891}$. A multiplication matrix has the irreducible factor

$$t^6 - 11t^5 + 5t^4 + 38t^3 - 5t^2 - 11t - 1,$$

and the certificate supplies two noncommuting automorphisms of its degree-six field. Their defining substitutions and nonzero commutator are checked by exact polynomial reduction. Thus this case is decided by the uniform arithmetic criterion without an induction search.

3.3 Positive dimensions after a tensor-sign change

Lemma 3.4 (Tensor-sign Galois reduction)**.** *Let $\mathcal{C}$ be spherical and let σ be an embedding of its simple-dimension field into $\mathbb{R}$. Put $\eta_i = \operatorname{sgn}(\sigma(\dim X_i))$. If*

$$\eta_0 = 1, \qquad N_{ij}^k > 0 \Longrightarrow \eta_i \eta_j = \eta_k, \tag{3.1}$$

then $\mathcal{C}$ has a pseudo-unitary Galois conjugate after changing its spherical structure by a tensor automorphism of the identity.

Proof. Extend σ to the field of definition and conjugate the categorical structure, as in the Galois reduction of [12]. The scalars η_i define a tensor automorphism of the identity precisely because of (3.1). They agree on duals and square to one, so twisting the pivotal structure preserves sphericality. The resulting dimensions are $|\sigma(\dim X_i)| > 0$. A positive ring dimension character is the Frobenius–Perron character, so this spherical structure is positive and the conjugate is pseudo-unitary. □

This lemma must not be replaced by arbitrary basis sign changes: the support condition (3.1) is essential. The spherical checker now tests it at every real embedding of a remaining character component. Combined with the complete signed-sphericalization tests of Section 3.4, it supplies general exclusion certificates for $R^{(6)}_{2,37}$, $R^{(6)}_{2,57}$ and $R^{(6)}_{3,204}$. The new step concerns spherical character components; the exhaustive signed analysis is still required before a conclusion is promoted to general nonexistence.

3.4 Signed sphericalization at rank six

Let $\mathcal{C}$ be an arbitrary fusion category. Its canonical sphericalization $\widetilde{\mathcal{C}}$ has a sign object ε of dimension -1 and two lifts of every simple object, interchanged by tensoring with ε; see [14, 36]. Choose a unit-preserving set of lifts. Passing to

$$T = K_0(\widetilde{\mathcal{C}})/(\varepsilon + 1)$$

gives integral signed structure constants satisfying

$$|T_{ij}^k| \le N_{ij}^k, \qquad T_{ij}^k \equiv N_{ij}^k \pmod 2. \tag{3.2}$$

They retain associativity, the unit, duality, and Frobenius reciprocity. All these statements concern a necessary quotient of an actual sphericalization; conversely, an admissible signed table need not have a categorification.

The sphericalization arguments in [12, 11] choose the lift of each self-dual object with positive second indicator. We keep this normalization. In particular, the square-channel congruences are modulo four rather than merely modulo two whenever that normalization applies. Non-self-dual lifts are chosen compatibly with duality. These restrictions are part of the cited normalization theorem, not heuristic reductions of a search.

3.4.1 An exhaustive oriented parameterization

Set $t_{abc} = T_{ab}^{c^*}$. In the absence of commutativity, identify only the orbits generated by

$$(a,b,c) \longmapsto (b,c,a), \qquad (a,b,c) \longmapsto (c^*,b^*,a^*). \tag{3.3}$$

These are the cyclic and dual-reflection consequences of reciprocity. An arbitrary transposition is not allowed. Unit and duality coefficients are fixed first. Give each remaining orbit its interval in (3.2), impose the applicable square-channel congruences, and substitute into all ordered associativity equations.

Proposition 3.5 (Exhaustive signed search)**.** *The finite search just described contains the signed quotient of every hypothetical categorification, after the indicated lift normalization. If all its associative tables are rejected by necessary signed sphericalization criteria, the original ring has no complex categorification.*

Proof. The quotient construction gives (3.2) and (3.3). Thus every quotient occurs in the finite product of integer domains. Every pruning operation is a necessary consequence of an equation over those domains. At a leaf the full ordered associativity identities are checked. Exhausting the product therefore exhausts every possible normalized quotient. A genuine category would supply one of the remaining tables and one of its spherical dimension characters, so rejection of all tables and all such characters is contradictory. □

The implementation checks machine-integer bounds before native computation and does not interpret overflow or timeout as exhaustion. The exact verifier reconstructs the parameter domains and the complete list of surviving signed tables; it does not merely recheck a collection of stored rejected examples.

3.4.2 When the signed algebra must be commutative

Let $d = (1, d_1, \dots, d_5)$ be the dimension character of the selected lifts, $K = \mathbb{Q}(d_1, \dots, d_5)$, and $\delta = \sum_i d_i d_{i^*}$. Inflating a representation of R to $K_0(\widetilde{\mathcal{C}})$ doubles its formal codegree. Since the sphericalization has dimension 2δ, central induction gives

$$\delta/f_E \in \mathcal{O}_K, \qquad f_E \in K \quad (E \in \mathrm{Irr}(R \otimes \mathbb{C})). \tag{3.4}$$

Indeed, [36, Theorem 2.13] gives a central simple of dimension δ/f_E, and its restriction is an integral combination of the selected lifts and their negatives. Its dimension is a nonzero algebraic integer in K, proving both assertions.

Proposition 3.6 (Rank-six commutative descent)**.** *Suppose R is commutative of rank six and has a hypothetical categorification. Its signed quotient $T \otimes \mathbb{C}$ is commutative if any of the following tensor-checkable conditions holds:*

(a) all six basis elements are self-dual;

(b) exactly two basis elements are self-dual;

(c) a formal codegree has degree greater than two;

(d) the formal codegrees generate two different quadratic subfields;

(e) zero has multiplicity less than four in $\det(tI - C_R) \bmod 2$.

Proof. The coefficient-of-unit pairing of a signed quotient is still positive definite on the distinguished basis after complex conjugation. Its regular matrices are adjoint under duality. Hence the algebra is semisimple. It has a one-dimensional dimension representation. If it is noncommutative of dimension six, its only possible Wedderburn type is

$$\mathbb{C} \oplus \mathbb{C} \oplus M_2(\mathbb{C}). \tag{3.5}$$

The Galois orbit of the dimension character has at most two elements, so $[K : \mathbb{Q}] \leq 2$. Equation (3.4) excludes (3.5) in cases (c) and (d).

In case (a), duality is the identity on the signed basis. Being also an anti-automorphism, it forces commutativity. For (b), consider duality as a complex-linear involutive anti-automorphism of (3.5). The spherical dimension character is fixed, so both one-dimensional blocks are fixed. On $M_2(\mathbb{C})$, an involutive anti-automorphism is of orthogonal or symplectic type, with linear trace 2 or -2. The total trace is therefore 4 or 0. In the distinguished basis it is the number of fixed basis elements, namely 2, a contradiction.

Finally the unique two-dimensional representation of (3.5) is Galois invariant. Its formal codegree is a rational algebraic integer, hence an integer. The regular Casimir of T has the even eigenvalue $2f_E$ with multiplicity four. Thus its characteristic polynomial is divisible by t^4 modulo two. But (3.2) implies $C_T \equiv C_R \pmod 2$, contradicting (e). □

This proposition is a reduction, not a noncategorification criterion by itself. Once its hypotheses are verified, the commuting-character methods of [12, 11] apply to every signed branch. Otherwise we use Proposition 3.5. If a noncommutative table of type (3.5) remains, its one-dimensional characters are computed as the common kernel of the matrix commutators. That space has dimension two. The two-dimensional block and its formal codegree are then recovered from the regular Casimir with the multiplicities in (3.5). Each decomposition identity is checked exactly.

3.5 A scoped literature obstruction at the first frontier

The general existence question and the premodular existence question must be kept separate. The following consequence of the complete rank-six premodular fusion-rule catalogue is useful for the latter question.

Proposition 3.7. *If a rank-six fusion ring admits a braided spherical categorification, then its Frobenius–Perron dimension is at most* 168.

Proof. Use the fusion-rule classification of Dong–Schopieray [10, Section 5.3, Figures 13–15, 17 and 19]. The integral representation-ring entries and their listed braided variants have dimension at most 168. The other entries belong to the families below. Write $[j]_\ell = \sin(j\pi/\ell)/\sin(\pi/\ell)$. The inequalities

$$\sin(\pi/\ell) \geq 2/\ell, \qquad |\sin(j\pi/\ell)| \leq 1, \qquad \sqrt{5} < 3, \qquad \sqrt{21} < 5$$

give the following rational upper bounds. Each is at most 168.

Family of fusion rules	Upper bound
Pointed rank three times Fibonacci	12
$\mathrm{Rep}(S_3)$ times Fibonacci	24
Dimension-eight Ising and generalized-Ising rules	8
Ising times Fibonacci	16
Pointed rank two times $\mathcal{C}(A_1, 7)_{\mathrm{ad}}$	51
Fibonacci times $\mathcal{C}(A_1, 7)_{\mathrm{ad}}$	102
$\mathcal{C}(G_2, 21)$	110
Dimension-twenty metaplectic rules	20
$\mathcal{C}(A_1, 13)_{\mathrm{ad}}$	2197/16
$\mathcal{C}(B_2, 9)_{\mathrm{ad}}$	81
$\mathcal{C}(A_1, 12)_{\mathrm{ad}}$	108

For example, the Fibonacci dimension is less than 4, and the sum of squared dimensions of $\mathcal{C}(A_1, 7)_{\mathrm{ad}}$ is at most $1 + 2(7/2)^2$. For the two adjoint A_1 entries in the last three rows, use their dimension formulae $\frac{13}{4}\csc^2(\pi/13)$ and $3\csc^2(\pi/12)$. For B_2, the formula is $9u^2$ with $u = 1 - 2\cos(8\pi/9)$ and $|u| \leq 3$. For G_2, its six dimensions are bounded by $1, 4, 4, 4, 5, 6$. These cover all rows of the cited catalogue. The caveat there about counting braided equivalence classes for two families does not add fusion rules to the list. □

The implementation derives the positive character and its number-field embedding from each tensor. It then decides $\mathrm{FPdim}(R) - 168 > 0$ by exact real-root isolation, recording the defining polynomial, dimension coordinates and isolating interval. No tensor identifier is used to select the outcome. In particular, the least-multiplicity residual $R^{(6)}_{3,215}$ has the automatically verified positive dimension vector

$$(1, 1, d, d, d-1, d+1), \qquad d = 4 + \sqrt{17}, \qquad \mathrm{FPdim}(R) = 136 + 32\sqrt{17} > 168.$$

It therefore has no premodular categorification. Its general, pivotal, spherical, pseudo-unitary and unitary existence questions are not resolved by this proposition, because none of those hypotheses alone supplies a braiding.

For a separate complete-tensor model audit, the rank-six ancillary data of Ng–Rowell–Wen [29] contain 192 modular records in 20 Galois orbits and 11 based-ring types. The supplied parser reads only literal integer tensors and compares all 120 permutations fixing the unit. Their largest fusion coefficient is 2, so this source supplies no new model at multiplicity eight or for the multiplicity-three frontier. This comparison is a model search within a specified literature class, not an exclusion of arbitrary fusion categories.

4 One central-induction certificate framework

The following subsections are blocks of one restriction-and-trace system. The Gram data and representation multiplicities are fixed first. Character labels, reciprocity, Galois orbits, indicators, and modular identities then add equations or remove impossible columns from that

same system. Each block states its own scope: Frobenius–Perron positivity is used only in the pseudo-unitary blocks, whereas the signed block retains negative spherical dimensions. A Farkas vector certifies infeasibility only after all low frames, character assignments, and permitted indicator choices have been exhausted. No particular block is mandatory when a shorter inherited certificate already proves the required exclusion.

4.1 Central induction with representation multiplicities

Assume in this section that $\mathcal{C}$ is pseudo-unitary, with its positive spherical structure. Let $F : Z(\mathcal{C}) \to \mathcal{C}$ be forgetful and I its adjoint. Put

$$B_{ji} = [FI(b_j) : b_i] = \sum_{a,k} N_{aj}^{k} N_{k,a^*}^{i}. \tag{4.1}$$

This is the induction Gram matrix. In a commutative ring it agrees with the regular Casimir. In a noncommutative ring one must use (4.1), not silently substitute the Casimir.

For a central simple Z, write $v_Z = ([F(Z) : b_i])_i$. Adjunction gives

$$B = \sum_{Z \in \operatorname{Irr} Z(\mathcal{C})} v_Z v_Z^{\mathsf{T}}. \tag{4.2}$$

The summands of $I(\mathbf{1})$ are indexed by irreducible representations E of $R \otimes \mathbb{C}$:

$$I(\mathbf{1}) = \bigoplus_E r_E A_E, \quad (v_{A_E})_0 = r_E, \quad \dim A_E = D/f_E, \quad \theta_{A_E} = 1. \tag{4.3}$$

These are the formal-codegree and induction statements of [36, Theorems 2.5,2.13].

Theorem 4.1 (Multiplicity-weighted low induction)**.** *For each E let $v_E \in \mathbb{Z}_{\geq 0}^{6}$ satisfy*

$$(v_E)_0 = r_E, \qquad d \cdot v_E = D/f_E, \qquad \sum_E r_E v_E = B_{0,*}. \tag{4.4}$$

A necessary condition for a pseudo-unitary categorification is that at least one such collection has a nonnegative integral Gram completion

$$Q = B - LL^{\mathsf{T}} = \sum_{Z \not\subset I(\mathbf{1})} v_Z v_Z^{\mathsf{T}}, \qquad L = (v_E)_E, \tag{4.5}$$

and satisfies the central-indicator necessary conditions of [12, 11]. In particular, Q is entrywise nonnegative and positive semidefinite. The columns of L occur once each in LL^{T}; they are not repeated r_E times.

For a noncommutative rank-six ring the low unit multiplicities are $1, 1, 2$. Equations (4.4) *and* (4.5)*, with these multiplicities, give a finite exhaustive low-column search.*

Proof. The first two identities of (4.4) are (4.3). Restricting $I(\mathbf{1})$ proves the third. Separating the distinct simples occurring in $I(\mathbf{1})$ from the sum (4.2) proves (4.5). This also explains why r_E weights the row-sum equation but not the number of copies of a column in L.

Finiteness follows from $(v_E)_i^2 \leq B_{ii}$ and $r_E(v_E)_i \leq B_{0i}$. Thus each entry belongs to an explicitly bounded integer interval. All dimension equalities are checked in the exact number field. The usual trace identities involve only restriction multiplicities and categorical dimensions, so their proofs and the finite relative-trace construction do not require commutativity of R once the correct low data and Gram matrix are supplied. Finally a noncommutative semisimple rank-six algebra with a dimension character has type (3.5), giving the stated multiplicities. □

In the implementation the one-dimensional character subspace is computed before selecting a Frobenius–Perron embedding. The unique matrix-block eigenvalue c of the Casimir gives $f_E = c/2$, and the identity

$$\det(tI - C_R) = (t - c)^4 \det(tI - C_R|_{\text{linear}})$$

is checked. A positive gate from Proposition 3.1(iv) is required before a pseudo-unitary exclusion is promoted to a general one. The five higher-multiplicity noncommutative Haagerup-type tensors in the census are excluded in this way. The multiplicity-one Haagerup ring and the group ring of S_3 are retained positive controls.

4.2 Balanced finite-indicator systems

The principal positive-dimension test is a single relative-trace construction. We strengthen it without introducing a new case-specific obstruction. Throughout this section suppose all dimensions lie in $K = \mathbb{Q}$ or $K = \mathbb{Q}(\sqrt{q})$, with $q > 1$ squarefree in the latter case. The low matrix L is one of the possibilities of Theorem 4.1.

4.2.1 The inherited finite system

For a simple b_i, the Ng–Schauenburg identity [30] is

$$\sum_Z (v_Z)_i \, \dim(Z) \, \theta_Z^k = D\nu_k(b_i). \tag{4.6}$$

The first indicator vanishes for nonunit simples, and the second is zero for non-self-dual simples and ± 1 for self-dual simples. Higher indicators satisfy the bounds used in the finite-indicator theorem of [11]. We use exactly that theorem, including its permitted indicator choices and rational inequalities, rather than assuming unknown indicators have prescribed values.

For a chosen height h, take joint normalized relative traces of $(\theta, \dots, \theta^h)$ to K. The exponent prime support is controlled by the categorical Cauchy theorem [5]. Let P be the prime divisors of the norm of D, and let f be the conductor of K (take $f = 1$ for $K = \mathbb{Q}$). The trace vectors are covered by roots whose orders divide

$$M_h = \prod_{p \in P} p^{\max\{1, v_p(f)\} + \lfloor \log_p h \rfloor}, \tag{4.7}$$

together with the zero vector. As proved in [11], a higher prime-power layer contributes zero to every one of the first h traces. Thus this is not a bound imposed on the actual twist orders. Ramanujan and quadratic Gauss sums compute the finite trace vectors exactly. Extra vectors may be retained as a relaxation; they cannot produce a false exclusion.

An admissible high column has zero unit entry, a nonnegative integral tail v, and $Q - vv^{\mathsf{T}} \succeq 0$. Entrywise Gram bounds and the algebraic-integrality condition $D/\dim(v) \in \mathcal{O}_K$ further restrict it. We also require $\dim(v)$ to be a d-number: the center is braided, so its simple squared norms are d-numbers by [35, Theorem 1.8]; in the positive spherical structure these are the squares of the dimensions, and [35, Corollary 2.9] removes the square. In a quadratic field this is the exact condition that the square of the trace divided by the norm is integral. Denote the finite list by $\mathcal{V}$, and the finite joint trace list by $\mathcal{T}$. Variables $x_{v,\tau} \geq 0$ give the number of high simples with column v and trace type τ; the linear relaxation allows nonintegral values. Gram identities, (4.6), and the cited higher-indicator inequalities give an integer linear system

$$Ax = b, \qquad x \geq 0, \tag{4.8}$$

after adding slack variables and clearing denominators. The code reconstructs this system from the tensor, the low matrix, and the indicator choices.

4.2.2 Duality and Galois balance

Write $v_i^* = v_{i^*}$. Low simples are stable under duality, so the high part is stable as well. Duality preserves the ribbon twist: $\theta_{Z^*} = \theta_Z$, not θ_Z^{-1}. Consequently the actual integer counts satisfy

$$x_{v,\tau} = x_{v^*,\tau} \qquad (v \in \mathcal{V},\ \tau \in \mathcal{T}). \tag{4.9}$$

If a partner column is absent from the list, its variable is zero.

For the quadratic case let $\sigma(\sqrt{q}) = -\sqrt{q}$ and define, for a positive central dimension $\beta \in K$,

$$\gamma(\beta) = \left| \frac{D}{\sigma(D)} \sigma(\beta) \right|, \tag{4.10}$$

where absolute value is at the fixed real embedding. Let c_β be the number of *distinct* low central simples of dimension β. The low multiplicity r_E in $I(\mathbf{1})$ is not included in this count.

Lemma 4.2 (Dimension-count balance)**.** *Galois symmetry permutes the central simples with dimension β onto those with dimension $\gamma(\beta)$. Hence their high counts satisfy*

$$\sum_{\dim(v)=\beta,\tau} x_{v,\tau} - \sum_{\dim(v)=\gamma(\beta),\tau} x_{v,\tau} = c_{\gamma(\beta)} - c_\beta. \tag{4.11}$$

Moreover $\gamma^2(\beta) = \beta$.

Proof. Extend σ to a cyclotomic field of the center. Its normalized S-matrix is $s = S/D$, since the center has global dimension D^2. Signed Galois symmetry gives

$$\sigma(s_{0Z}) = \epsilon_Z s_{0,\widehat{\sigma}(Z)}, \qquad \epsilon_Z \in \{1, -1\}.$$

The entry s_{0Z} equals $\dim(Z)/D$. All central dimensions are positive. Taking absolute values therefore gives $\dim \widehat{\sigma}(Z) = \gamma(\dim Z)$, proving the permutation statement. Subtract the known low counts to obtain (4.11). Finally the map $u \mapsto (D/\sigma D)\sigma(u)$ squares to the identity. Taking absolute values commutes with changing a real sign, so the positive version squares to the identity as well. See also the squared-dimension form of Galois symmetry in [28, Equation (2.4)]. □

The factor $D/\sigma(D)$ and the absolute value in (4.10) are both essential. Omitting either does not give a valid positive dimension permutation in general.

Theorem 4.3 (Balanced finite-indicator criterion)**.** *Append* (4.9) *and, when quadratic,* (4.11) *to the finite system of [11]. If for every admissible low matrix and every allowed second-indicator choice the resulting system has an exact Farkas certificate*

$$A^{\mathsf{T}} y \geq 0, \qquad b^{\mathsf{T}} y < 0, \tag{4.12}$$

then there is no pseudo-unitary categorification. Coupled with a valid general-to-positive reduction, this excludes general complex categorification.

At a fixed height and with the same admissible low and high columns, this criterion contains the former unbalanced finite-indicator test: every old separating functional extends by zero on the appended rows.

Proof. An actual category supplies integral nonnegative counts satisfying the inherited system and both new balances, by duality and Lemma 4.2. But (4.12) would imply $0 > b^{\mathsf{T}} y = x^{\mathsf{T}} A^{\mathsf{T}} y \geq 0$. The last assertion follows by appending zero coordinates to y. □

This containment justifies using the balanced theorem as the common finite-indicator framework. It does not imply that every earlier non-indicator criterion is redundant. In particular, the cheap arithmetic filters and signed reductions remain useful.

4.2.3 An integral refinement and its scope

For a high-column list $v_1, \dots, v_s$, enumerate nonnegative integers m_j with

$$Q = \sum_{j=1}^{s} m_j v_j v_j^{\mathsf{T}}. \tag{4.13}$$

This is finite because each nonzero v_j has a positive diagonal entry. For each frame append $\sum_\tau x_{v_j,\tau} = m_j$ to the balanced system. Exhaustive infeasibility for every frame would be a further valid obstruction. We implement this refinement principally to diagnose the frontier. The feasible point in Section 7.2 survives it. We do not count this unproductive refinement as an independent successful exclusion criterion.

4.3 Cyclic-character compatibility for tensor powers

The preceding indicator system bounds different higher indicators separately. There is additional compatibility: several of them are traces of powers of a single finite-order linear operator. This gives extra linear equations inside the same exact-certificate framework.

Lemma 4.4 (Block rotation)**.** *In a pivotal fusion category, let $E_X^{(n)}$ be cyclic rotation on* $\mathrm{Hom}(1, X^{\otimes n})$*. For every positive divisor r of n,*

$$(E_X^{(n)})^n = \mathrm{id}, \qquad \mathrm{Tr}((E_X^{(n)})^r) = \nu_{n/r}(X^{\otimes r}) = \sum_i [X^{\otimes r} : X_i]\nu_{n/r}(X_i). \tag{4.14}$$

Here the trace is the ordinary trace on the displayed Hom space.

Proof. Pivotal strictification preserves the rotation operators. In the strict pivotal setting, equation (5.4) in [31] states that consecutive rotations of adjacent tensor blocks equal the rotation of their tensor product. Induction gives $(E_X^{(n)})^r = E_{X^{\otimes r}, X^{\otimes(n-r)}}$, exactly the identity used in the proof of [31, Theorem 5.1]. If $r \mid n$, group the factors into n/r consecutive blocks of length r. This identifies the operator with one rotation on the blocks. Taking traces gives the first indicator equality; the final equality is additivity of higher indicators on a direct sum. For $r = n$ it is the identity operator, with trace $[X^{\otimes n} : 1]$. □

Fix the dimension field K, the low columns, and the second indicators as in Section 4.2. For a positive integer n, let $\mathcal{U}_{K,n}$ be the finite set of vectors

$$(P_K(\zeta^r))_{r|n,\ r<n}, \qquad \zeta^n = 1,$$

where P_K is normalized relative trace from a common cyclotomic extension to K. This definition is independent of enlarging the common extension. The vectors are computed by the same Ramanujan and quadratic Gauss sums used for the center twists, but their orders now divide n exactly. In particular no unbounded-order approximation occurs in this auxiliary set.

Theorem 4.5 (Joint tensor-rotation criterion)**.** *In a pseudo-unitary categorification with simple dimensions in K, for each simple X and each $3 \le n \le h$ there are nonnegative integers $y_{X,n,u}$, $u \in \mathcal{U}_{K,n}$, such that*

$$\sum_u y_{X,n,u} = [X^{\otimes n} : 1], \tag{4.15}$$

$$\sum_u y_{X,n,u} u_r = \sum_i [X^{\otimes r} : X_i] P_K(\nu_{n/r}(X_i)) \qquad (r \mid n,\ r < n). \tag{4.16}$$

Together with the center trace identities, these are linear equations in the existing high-center counts and the new y variables after multiplication by D and expansion in a rational basis

of K. Infeasibility of the nonnegative real relaxation is an obstruction to pseudo-unitary categorification. Combined with any proved sphericalization or pseudo-unitarity reduction, it is a general categorification obstruction.

Proof. The operator in Lemma 4.4 is diagonalizable over $\mathbb{C}$ and has nth roots of unity as eigenvalues. Group its eigenvalues, with multiplicity, by their joint relative-trace vectors. This gives nonnegative integer counts satisfying (4.15) and (4.16). Use (4.6) to express each indicator on the right. Its denominator is the known scalar D, while its numerator is the known low contribution plus a linear combination of the high-center variables. Since P_K is K-linear, multiplying by D gives linear equations with coefficients in K. Clearing rational denominators gives an integer system. An integer solution would also be a nonnegative real solution; an exact separating functional therefore proves the claimed obstruction. □

For clarity, let ℓ_i be the dimension of the known low part of $I(X_i)$, with $\ell_0 = D$, and let (v, τ) index a high variable. The equation appended for a divisor r is exactly

$$D\sum_u y_{X,n,u}u_r - \sum_{v,\tau} x_{v,\tau}\Big(\sum_{i>0}[X^{\otimes r} : X_i]v_i\Big)\dim(v)\tau_{n/r} = \sum_i [X^{\otimes r} : X_i]\ell_i. \tag{4.17}$$

This formula specifies the implemented rows without a graphical normalization ambiguity. The unit indicator is one, and unit-free high columns have unit entry zero.

The criterion strengthens the existing system by adding equations and variables with nonnegative constraints; no previously valid inequality is dropped. Old dual certificates extend by zero. Its height-six application excludes $R^{(6)}_{2,55}$, the self-dual Klein-four quadratic ring with two noninvertibles of dimension $2+\sqrt{6}$. The complete ten low-column alternatives and every permitted second-indicator choice are checked. The two-Frobenius–Perron-dimension reduction supplies the general scope. No separate theorem declaring this particular ring impossible is used by the program.

The cyclic-four ring $R^{(6)}_{2,4}$ has the same dimension type but does not satisfy that exclusion certificate. Section 7.2 records an exact integer-count feasible point for the stronger system. Thus neither dimension-type identification nor a failed numerical optimization can replace full-tensor testing.

4.4 Character-aware central induction

The dimension-only induction matrix forgets an important part of the formal-codegree correspondence: each of its low columns has a specified *character*, not only a specified dimension. Retaining these characters adds exact modular reciprocity and a joint Galois-spectrum condition to the same finite-indicator system. No classification of modular data, numerical eigenvalue decision, or enumeration of center fusion rings is required.

4.4.1 The character attached to a low column

Let $\mathcal{C}$ be pseudo-unitary, let $R = K_0(\mathcal{C})$ be commutative of rank n, and write $d_i = \mathrm{FPdim}(X_i)$ and $D = \sum_i d_i^2$. For a character $\chi : R \to \mathbb{C}$, put

$$f_\chi = \sum_i \chi(X_i)\chi(X_i^*), \qquad \beta_\chi = D/f_\chi.$$

Use the unnormalized modular matrix S of $Z(\mathcal{C})$, so that $S_{1W} = \dim W$. The correspondence preceding [36, Theorem 2.13] identifies the simple summand Z_χ of $I(1)$ by the equality of characters

$$\chi(FW) = \frac{S_{Z_\chi,W}}{\beta_\chi} \qquad (W \in \mathrm{Irr}\, Z(\mathcal{C})). \tag{4.18}$$

Indeed, pull back χ along F. The resulting character of the modular fusion ring is its unique normalized S-row, and [36, Theorem 2.13] identifies the row label, its dimension β_χ, and its multiplicity one in $I(1)$. Thus this statement uses the character-level correspondence, not an arbitrary choice among central simples of equal dimension.

For an integral restriction vector $v = (v_i)$ write $d(v) = \sum_i v_i d_i$ and $\chi(v) = \sum_i v_i \chi(X_i)$. If a_χ is the restriction vector of Z_χ, symmetry of S gives

$$\boxed{\beta_\chi \chi(a_\psi) = \beta_\psi \psi(a_\chi)}. \tag{4.19}$$

The unit column is labelled by the Frobenius–Perron character. When formal codegrees repeat, every dimension-compatible assignment of the remaining distinct characters must be considered. Selecting one assignment without exhausting the others would not be an obstruction.

4.4.2 Joint Galois closure

The next elementary spectral observation is useful well beyond the present induction problem.

Lemma 4.6 (Joint spectral closure)**.** *Let B be a commutative fusion ring and $b_1, \dots, b_s \in B$. The multiset*

$$\{(\varphi(b_1), \dots, \varphi(b_s)) : \varphi \in \mathrm{Irr}(B \otimes \overline{\mathbb{Q}})\}$$

is invariant under every automorphism of $\overline{\mathbb{Q}}$ over $\mathbb{Q}$. Consequently a finite set containing all these tuples must contain every Galois conjugate of each tuple that actually occurs.

Proof. The multiplication equations of B have integral coefficients. Composing a character with a field automorphism gives another character, and the inverse automorphism makes this operation a permutation. Equivalently, the commuting integer multiplication matrices have a Galois-invariant joint spectrum. Restriction to the specified elements preserves the multiset, including repeated restricted tuples. □

For a nonzero prospective restriction vector v define

$$\Lambda(v) = \left(\frac{\beta_\chi \chi(v)}{d(v)} \right)_{\chi \in \mathrm{Irr}\, R}. \tag{4.20}$$

If $v = FW$ for a central simple W, then (4.18) and symmetry give

$$\frac{\beta_\chi \chi(v)}{d(v)} = \frac{S_{W,Z_\chi}}{\dim W}.$$

These are the joint eigenvalues of the integer fusion matrices of Z_χ evaluated at the character labelled by W. In particular, Lemma 4.6 applies to the *whole tuple*. Checking its coordinates separately would be weaker.

Fix a candidate low matrix L. Let $\mathcal{H}_L$ be any finite set containing all possible high restriction columns for that low matrix; the Gram and algebraic-integrality construction in Section 3.1 and the finite-indicator implementation supplies such a set. Let $\mathcal{A}_L$ be its union with the low columns, and set

$$\mathcal{T}_L = \{\Lambda(v) : v \in \mathcal{A}_L\}.$$

Choose a finite Galois field K containing all character values and all d_i. Every coordinate of every tuple belongs to K. A column v is *forbidden* if $\sigma(\Lambda(v)) \notin \mathcal{T}_L$ for some $\sigma \in \mathrm{Gal}(K/\mathbb{Q})$. A forbidden low column rejects L; a forbidden high column must have multiplicity zero. Notice that (4.20) uses the fixed character order and D/f_χ, so this test is independent of the permutation used to label equal-dimensional low columns.

Theorem 4.7 (Character-aware finite-indicator obstruction)**.** *For every low matrix of a pseudo-unitary categorification with commutative Grothendieck ring:*

(i) there is a dimension-compatible bijection from its columns to $\operatorname{Irr} R$ *satisfying* (4.19)*;*

(ii) none of its low columns is forbidden by the joint Galois test;

(iii) the finite-indicator system remains feasible after all variables supported on forbidden high columns are set to zero.

Therefore the absence of an assignment in (i), failure of (ii), or an exact Farkas obstruction to (iii), for every candidate low matrix and every permitted second-indicator choice, excludes a pseudo-unitary categorification.

Proof. The formal-codegree correspondence proves (i). Every actual central restriction belongs to $\mathcal{A}_L$. Lemma 4.6, applied to the low central simples, proves (ii) and the asserted zero multiplicities. The finite-indicator variables group actual central simples by restriction and joint relative twist trace. Every variable with a forbidden restriction is therefore zero; the remaining actual multiplicities satisfy the inherited equations and inequalities. After slack variables are inserted these have the form $Ax = b$, $x \geq 0$. A rational vector y with $A^{\mathsf{T}}y \geq 0$ and $b^{\mathsf{T}}y < 0$ is impossible, because $b^{\mathsf{T}}y = x^{\mathsf{T}}A^{\mathsf{T}}y \geq 0$. All ordinary low-matrix rejections remain applicable before these additional tests. □

This is a strengthening of the existing finite-indicator framework, not a competing list of special forbidden rings. The implementation uses exact splitting fields of degree at most sixteen generated by quadratic eigenvalues of one or two ring generators. This is an implementation eligibility bound, not a hypothesis of the theorem. Every proposed character is checked against all multiplication equations; the complete set has cardinal n. Field automorphisms are constructed by splitting the primitive-element polynomial over K and checking that all of its distinct roots have been obtained. The verifier reconstructs all labellings, all possible high columns, the forbidden set, and every rational separating vector. It never calls the numerical optimizer used to discover a vector.

4.4.3 General categorification and the multiplicity-two completion

Two illustrative tensors give a compact reproducibility summary:

Ring	Low matrices	Reciprocity survivors	General reduction
$R^{(6)}_{2,20}$	77	2	two FP dimensions
$R^{(6)}_{2,41}$	84	2	two exhaustive signed tables

For $R^{(6)}_{2,20}$, both surviving matrices and both permitted second indicators have exact height-six obstructions. The two-FP-dimension Galois reduction of Proposition 3.1 makes this a general exclusion. For $R^{(6)}_{2,41}$, one reciprocity survivor is already rejected by the inherited height-six equations. For the other, there are 104 possible high columns; joint Galois closure forbids 66, and the restricted height-six system has an exact separating vector.

For this latter tensor the complete oriented signed enumeration has exactly two tables. Both are sign gauges of the original tensor, with sign vectors $(1, -1, -1, 1, 1, 1)$ and $(1, 1, 1, 1, 1, 1)$ in census order. The ordinary spherical-character test excludes every character outside the FP Galois orbit by a zero dimension or unequal dimensions on a dual pair. That orbit is excluded by Theorem 4.7. The signed-sphericalization reduction already established in the preceding sections therefore excludes *general* categorification, without assuming the conjectural existence of pivotal structures. These statements are recomputed from the tensors in the certificate, including the completeness of the oriented enumeration.

Table 2 records 23 certified examples of this common character-aware mechanism. Their stored proof trees include the complete sphericalization or Galois reduction required for general nonexistence. In particular, the signed calculation for $R^{(6)}_{3,132}$ is detailed in Section 4.4.4. The certificate controls recompute these applications without a numerical optimizer. This illustrative collection is not a separate allocation of the entire census: the current sequential counts, eligibility outcomes and bounded-search diagnostics appear in Section 7.1 and its machine-readable ledger.

Exact multiplicity	Example certificates
2	2
3	2
4	6
5	3
6	10
Total	23

Table 2: Illustrative character-aware induction certificates for Theorem 4.7, generated from the supplied exact proof examples. These are examples of the mechanism, not a disjoint allocation of the current census.

The multiplicity-three applications are detailed in Section 4.4.4. The complete consecutive classification range remains multiplicity two. In particular the counting bound eight is not presented as a categorification bound.

4.4.4 Modular equations retained inside the indicator system

We describe one further strengthening of Theorem 4.7. It uses the same high-column and relative-trace variables; it does not require enumerating an unknown center fusion ring. Its purpose here is also to specify exactly which stronger necessary conditions the remaining multiplicity-three example is known to satisfy.

Assume that all ring characters and positive dimensions lie in a real field K of degree at most two, and let P be the normalized relative trace from a common cyclotomic field to K. For a central simple W write $v_W = FW$, $d_W = d(v_W)$, and

$$H_{aW} = S_{Z_a,W} = \beta_a \chi_a(v_W).$$

Here Z_a ranges over the labelled low objects. Thus H_{aW} denotes an entry of the unnormalized matrix S.

Proposition 4.8 (Joint dimension, twist and low modular constraints)**.** *In addition to the preceding induction conditions, every pseudo-unitary categorification satisfies the following statements.*

Every coordinate of $\Lambda(v_W)$ is an algebraic integer. Under a Galois automorphism σ, the pair consisting of a simple dimension and its low fusion-eigenvalue tuple is permuted according to

$$(d_W, \Lambda(v_W)) \longmapsto \left(\left| \frac{D}{\sigma(D)} \sigma(d_W) \right|, \sigma(\Lambda(v_W)) \right). \tag{4.21}$$

This permutation preserves every joint relative twist-trace type

$$(P(\theta_W), P(\theta_W^2), \ldots, P(\theta_W^h)).$$

Finally, for any two low labels a, b,

$$\sum_W P(\theta_W) H_{aW} H_{bW} = D S_{Z_a, Z_b}. \tag{4.22}$$

Thus the finite indicator system may impose equality of multiplicities within each joint orbit in (4.21)*, separately for each trace type, and may impose all equations* (4.22)*. A missing orbit member forbids a candidate column; an exact infeasibility certificate for the enlarged system is a pseudo-unitary obstruction.*

Proof. The entries of $\Lambda(v_W)$ are eigenvalues of integer fusion matrices, so they are algebraic integers. Use the canonical normalized modular representation of the center, namely S/D and T. Its Galois symmetry gives

$$\sigma(S_{0W}/D) = \epsilon_\sigma(W) S_{0,\widehat{\sigma}W}/D, \qquad \Lambda(v_{\widehat{\sigma}W}) = \sigma(\Lambda(v_W)),$$

where $\epsilon_\sigma(W) \in \{1, -1\}$; see [5, equations (2.12)–(2.13)]. Positivity of all simple Frobenius–Perron dimensions gives exactly (4.21), including its absolute value.

For this canonical center normalization, the Galois symmetry of T is $\theta_{\widehat{\sigma}W} = \sigma^2(\theta_W)$, by [5, Theorem 2.9(b)]. A square of a Galois automorphism fixes K, since its degree is at most two. The normalized trace P is invariant under automorphisms over K, and cyclotomic Galois groups are abelian. Consequently $P(\theta^j_{\widehat{\sigma}W}) = P(\theta^j_W)$ for every j. The simultaneous dimension-and-eigenvalue permutation therefore gives multiplicity balances within each trace type, including the low objects in the twist-one type. This statement concerns multiplicities of tuples; it does not assert that a restriction vector itself is Galois invariant.

The positive Gauss sum of the center is D. Its modular relation is

$$D S_{Z_a, Z_b} = \theta_{Z_a} \theta_{Z_b} \sum_W \theta_W S_{Z_a, W} S_{W, Z_b},$$

by [5, equation (2.6)]. Both low twists are one. All the other coefficients in this equation belong to K, by the character formula. Applying P proves (4.22). The actual central simples satisfy the resulting rational equations and nonnegative multiplicity constraints. An exact Farkas obstruction therefore applies as in Theorem 4.7. □

The code reconstructs all finite Galois conjugates, checks algebraic integrality by integer characteristic polynomials, and adds coefficient rows over $\mathbb{Q}$. This is one refinement of the common induction system, not an independently allocated family of forbidden rings. In particular, the final general exclusion of $R^{(6)}_{3,132}$ already has a certificate using Theorem 4.7; no stronger criterion is credited when an inherited one suffices.

4.4.5 Exact multiplicity-three outcome and the higher scan

The general certificate for $R^{(6)}_{3,132}$ has the following finite structure. Its positive type is $(1, 1, d, d+1, d+1, d)$, with $d = 4 + \sqrt{21}$, and its global dimension is $168 + 36\sqrt{21}$. There are 172 low matrices. Modular reciprocity rejects 168; the other four have exact height-six indicator certificates after the joint-spectrum exclusions. The general sphericalization reduction exhausts sixteen oriented signed tables. Eight are sign gauges of the original table; for the other eight, all character branches are rejected by the checked norm or zero-dimension tests. These are certificate statistics reconstructed from the full tensor. They are not premises selected by its identifier.

Corollary 6.12 supplies the unitary model for $R^{(6)}_{3,218}$. Thus the only remaining exact multiplicity-three tensor is $R^{(6)}_{3,215}$. The strengthened rational system above has an exact feasible

point for that tensor: 230 equations in 199 nonnegative integer variables, including all indicator-slack coordinates, and formal central rank 22. Its low frame, character labelling, second indicators and all rational coordinates are stored in `systems/integral_frontier_215.json`. The verifier reconstructs the equations and checks the equalities exactly. The point is neither a modular datum nor a center fusion ring: individual twists and a complete associative fusion tensor have not been supplied. In particular, it does not prove categorifiability or noncategorifiability.

The common framework is applied at every multiplicity in the extended census. Structure-specific pseudo-unitary certificates remain separate from general exclusions. In particular the non-positive branches of $R^{(6)}_{4,375}$ and $R^{(6)}_{4,446}$ are closed by Section 4.5; a positive-dimension obstruction alone would not suffice. The surviving pseudo-unitary-only exclusions are $R^{(6)}_{4,162}$, $R^{(6)}_{7,3582}$ and $R^{(6)}_{7,3812}$. The complete general ledger and its distinct structure fields, rather than historical bounded-search outcomes, state the classification.

4.5 The dimension order and signed central traces

The positive-dimension induction system has two extensions that are useful when an exhaustive sphericalization search leaves a non-positive spherical character. The first sharpens an arithmetic condition without adding any central variables. The second retains the algebraic part of the same trace system, but discards every inequality that depends on positivity of categorical dimensions.

4.5.1 The actual order of simple dimensions

Let $d = (1, d_1, \ldots, d_{r-1})$ be a dimension character under consideration, and let $K = \mathbb{Q}(d_i)$ and $\delta = \sum_i d_i d_{i^*}$. The finitely generated group

$$\mathcal{O}_d = \sum_i \mathbb{Z} d_i \subset K$$

is an order: it contains 1, consists of algebraic integers, and is closed under multiplication because the d_i satisfy the multiplication equations with integer coefficients. It need not be the maximal order of K.

Proposition 4.9 (Dimension-order obstruction)**.** *If a fusion ring R has a spherical categorification with simple dimensions d_i, then for every formal codegree f of R,*

$$f \in K, \qquad \delta/f \in \mathcal{O}_d. \tag{4.23}$$

The same conclusion holds for the dimensions of selected lifts in the canonical sphericalization of an arbitrary categorification of R. Thus the order-membership test may be applied to each dimension-character branch of a complete signed-sphericalization search.

Proof. In the spherical case the central simple A_E attached to an irreducible representation of R with codegree f has dimension δ/f, by [36, Theorem 2.13]. Its forgetful image is an object of the original category. Its dimension is therefore a nonnegative integral linear combination of the d_i, in particular an element of $\mathcal{O}_d$. Since δ is nonzero, this also puts f in K.

For an arbitrary categorification use its sphericalization, of dimension 2δ. An irreducible representation of R lifts through the forgetful quotient to one with formal codegree $2f$; both selected and opposite basis lifts contribute to that codegree. The same central induction theorem gives dimension $(2\delta)/(2f) = \delta/f$. All simple dimensions of the sphericalization are among $\{d_i, -d_i\}$, so forgetting this central object gives an integral linear combination of the selected dimensions. This proves the assertion without assuming that the selected lifts themselves form a fusion subcategory. □

The test is finite and exact. Express the d_i in a rational power basis of K, clear one common denominator, and put their coordinate lattice in Hermite normal form. Membership of δ/f is then an integer linear-algebra question. A negative certificate can equivalently give a rational linear functional $\ell : K \to \mathbb{Q}$ satisfying

$$\ell(d_i) \in \mathbb{Z} \quad \text{for all } i, \qquad \ell(\delta/f) \notin \mathbb{Z}.$$

The checker verifies the defining polynomial of f, reconstructs the lattice and the functional, and checks both displayed conditions. A failure on one character component is not a noncategorification proof unless all possible sphericalization and character branches are closed. In particular a failed character of a positively realized ring cannot be promoted to an exclusion of that ring.

For illustration, the complete certificate for $R^{(6)}_{4,375}$ has four oriented signed tables, all gauges of the original tensor. Its nonzero spherical components have order $\mathbb{Z}[\sqrt{5}]$ and dimensions

$$\delta = 70 + 30\sqrt{5} \quad \text{or} \quad \delta = 30 - 10\sqrt{5}.$$

The codegree 4 gives quotients outside that order in both cases. The supplied functional certificates, not this identifier or this displayed calculation, are the inputs to the automated verification.

4.5.2 The algebraic trace system without positive dimensions

Suppose a spherical character takes values in a real field K of degree at most two, has nonzero simple dimensions, and satisfies $d_i = d_{i^*}$. Write $\delta = \sum_i d_i^2$. The restriction vectors of central simples remain nonnegative integral vectors, even when some categorical dimensions are negative. Consequently the Gram identity (4.2), its finite entry bounds, and the formal-codegree low-column equalities remain valid. They do not use an inequality between categorical dimensions.

For a commutative input let χ_a be a character label of a low simple Z_a and put $\beta_a = \delta/f_a$. If v_W is a candidate restriction vector, define

$$d_W = d \cdot v_W, \qquad H_{aW} = \beta_a \chi_a(v_W), \qquad \Lambda_W = (H_{aW}/d_W)_a.$$

Assume that the full ring character table is defined over K. The character correspondence of Theorem 4.7 gives $H_{aW} = S_{Z_a,W}$, with S unnormalized. The following formulation uses no absolute value of d_W.

Theorem 4.10 (Signed central-induction obstruction)**.** *A spherical categorification with the specified character supplies a nonnegative integral solution of the following finite necessary system. The unknowns count high central restriction vectors and their joint relative twist-trace types. They satisfy the Gram equations, the signed identities*

$$\sum_W (v_W)_i d_W \theta_W^j = \delta \nu_j(b_i), \tag{4.24}$$

and the cyclic-operator bounds on the right-hand indicators. Low columns have twist one and dimension β_a. They admit at least one complete character labelling satisfying modular reciprocity.

All coordinates of Λ_W are algebraic integers. Their joint Galois types have balanced multiplicities, separately for each relative twist-trace type. An absent Galois type forbids the corresponding restriction vector. Finally, if P is the normalized trace from a common cyclotomic field to K, then

$$\sum_W P(\theta_W) H_{aW} H_{bW} = \delta S_{Z_a,Z_b} \tag{4.25}$$

for every pair of low labels.

Exhaustive failure of the system for all bounded low columns, all compatible character labellings, and all permitted second indicators excludes the specified spherical character. Exact rational Farkas certificates suffice for failure of the nonnegative rational relaxation.

Proof. Adjunction proves the Gram identity with nonnegative restriction multiplicities. Central induction supplies the low dimensions and their unit multiplicities exactly as in (4.3); neither statement assumes pseudo-unitarity. The trace formula [30] proves (4.24) with the actual, possibly negative, dimensions. Bounds obtained as traces of finite-order cyclic operators on ordinary complex vector spaces do not assume positivity of categorical dimensions. Only these algebraic bounds, and not the positive low-mass inequalities, are retained here.

The finite relative-trace list is justified by the exponent-prime support theorem [5] and the same relative-trace calculation used in (4.7), with δ in place of D. The high columns are bounded by the positive integral Gram matrix, not by the sign of d_W. Their additional algebraic-integrality and d-number conditions follow from the spherical modular center as in the proof of Theorem 4.3. Columns of dimension zero are impossible.

Each Λ_W is the restriction of the simultaneous character of integer fusion matrices of the center to the labelled low objects. It therefore consists of algebraic integers, and its full Galois orbit occurs with the corresponding multiplicities. For the normalized center representation $S/\delta, T$, the permutation of columns sends θ_W to $\sigma^2(\theta_W)$; see [5, Theorem 2.9]. A square fixes the real field K of degree at most two, so normalized relative traces of every twist power are unchanged. This proves the asserted balances of joint eigenvalue types. It does not identify d_W with its absolute value, nor impose the positive dimension-orbit rule (4.21).

The center has Gauss sum δ. Its modular relation, together with the twist-one property of low objects, is $\sum_W \theta_W S_{Z_a,W} S_{W,Z_b} = \delta S_{Z_a,Z_b}$. All coefficients other than the twists lie in K. Taking P proves (4.25). The actual central simples give integer counts satisfying all the displayed equations. Expanding them in a rational basis of K and relaxing to nonnegative rational counts can only enlarge the solution set. Farkas' lemma therefore proves the last assertion. □

Exact scope of the implementation. The routine first checks the complete signed character table, its multiplication equations, duality, nonzero dimensions and field degree. It enumerates low columns using Gram bounds and exact dimension equalities, without calling a positive-dimension low-mass test. It then uses signed versions of the inherited indicator rows and adds joint spectrum and low modular rows. In particular it does not reuse the absolute-value dimension balances of Proposition 4.8. Only the second-indicator rows depend on the chosen signs, and each such row depends on one sign. The checker reconstructs the all-minus and all-plus endpoint systems and selects the exact normalized row for each sign choice. Independent mixed-sign reconstructions check this optimization; every separating-vector inequality is still verified. The second indicators of self-dual objects are both allowed unless an independently checked condition forces their sign.

For $R^{(6)}_{4,446}$ the remaining non-positive component is

$$d = (1, 1, -1, -1, 2 - \sqrt{6}, 2 + \sqrt{6}), \qquad \delta = 24.$$

There are 290 low matrices. Reciprocity leaves two, each with two character labellings; all 32 choices of second indicators are retained for each labelling. The 128 exact separation certificates exclude these systems. The positive character component has the preceding character-induction obstruction. The full oriented search has sixteen gauge tables, so the assembled certificate excludes general categorification, not just this displayed spherical structure.

4.5.3 Application and structure-specific conclusions

Component-level rejection and general nonexistence are different conclusions. For example, rejecting one character of $R^{(6)}_{4,174}$ does not exclude that ring. Complete proof trees for the order and signed framework include

$$R^{(6)}_{4,375}, \quad R^{(6)}_{4,446}, \quad R^{(6)}_{5,1507}, \quad R^{(6)}_{5,1508}, \quad R^{(6)}_{5,1582}, \quad R^{(6)}_{6,2767}, \quad R^{(6)}_{7,1907}.$$

For the last five, shorter norm and moment certificates close the exhaustive signed alternatives, and these are retained rather than credited to a stronger criterion unnecessarily. The finite applications are specified by full tensors and exact proof trees.

The separate certificates for $R^{(6)}_{4,162}$, $R^{(6)}_{7,3582}$ and $R^{(6)}_{7,3812}$ exclude pseudo-unitary and therefore unitary categorification only. Their general, pivotal and spherical existence fields remain unknown. This distinction is enforced by the ledger checker, not left to a reader's interpretation of a single status label.

4.6 Joint products in the first unresolved center

The first general residual is still $R^{(6)}_{3,215}$. Its positive dimension vector, computed from its multiplication tensor, is

$$(1, 1, d, d, d-1, d+1), \qquad d = 4 + \sqrt{17}, \qquad D = \mathrm{FPdim}(R) = 136 + 32\sqrt{17}.$$

A feasible induction system is not a categorification. We strengthen that system by requiring all products of its distinguished central objects to share one integral Gram decomposition. This retains fourth-order modular moments as well as the restriction equations, and is more restrictive than testing each product independently.

Proposition 4.11 (Joint low-product Gram condition)**.** *Suppose that $\mathcal{C}$ is pseudo-unitary. Let $Z_0 = \mathbf{1}, Z_1, \dots, Z_{s-1}$ be distinct self-dual simple objects of $\mathcal{Z}(\mathcal{C})$ whose restrictions $v_a = [F(Z_a)]$ are known. Write $D = \mathrm{FPdim}(\mathcal{C})$, $d_W = \mathrm{FPdim}(W)$ and $\lambda_a(W) = S_{aW}/d_W$, using the unnormalized center matrix S. For $p = (a, b)$, $1 \le a \le b < s$, put*

$$t_{p,e} = \frac{1}{D^2} \sum_W d_W^2 \lambda_a(W)\lambda_b(W)\lambda_e(W), \qquad 0 \le e < s,$$

where $\lambda_0(W) = 1$. Define

$$Q_{(a,b),(c,d)} = \frac{1}{D^2} \sum_W d_W^2 \lambda_a(W)\lambda_b(W)\lambda_c(W)\lambda_d(W) - \sum_{e=0}^{s-1} t_{(a,b),e} t_{(c,d),e}.$$

There is a nonnegative integral matrix $B = (b_{pW})$, with columns indexed by the remaining simple central objects, such that

$$Q = BB^{\mathsf{T}}, \qquad \sum_{W \notin \{Z_0, \dots, Z_{s-1}\}} b_{pW}[F(W)] = v_a v_b - \sum_{e=0}^{s-1} t_{p,e} v_e. \tag{4.26}$$

In particular every $t_{p,e}$ and every entry of Q is a nonnegative integer, and Q is positive semidefinite.

Proof. The Verlinde formula gives $t_{p,e} = [Z_a Z_b : Z_e]$. Self-duality makes these rows of S real and permits the displayed formulas without conjugations. Orthogonality of the normalized matrix S/D gives

$$[Z_a Z_b, Z_c Z_d] = D^{-2} \sum_W d_W^2 \lambda_a(W)\lambda_b(W)\lambda_c(W)\lambda_d(W).$$

Subtracting the contributions of the specified low objects leaves $Q_{pq} = \sum_W [Z_a Z_b : W][Z_c Z_d : W]$, where the sum is over the remaining objects. Set $b_{pW} = [Z_a Z_b : W]$ and apply the tensor functor F to each product. This proves both identities in (4.26). □

This condition has a useful inexpensive arithmetic refinement. Group high objects by their identical restriction vectors $v_1, \dots, v_h$, and let $c_j > 0$ be the numbers of objects in those types. If the v_j are linearly independent over $\mathbb{Q}$, the restriction equation determines uniquely the type totals $T_{pj} = \sum_{W:F(W)=v_j} b_{pW}$. The totals must be nonnegative integers. With $T_{pj} = c_j q_{pj} + r_{pj}$, $0 \le r_{pj} < c_j$, necessarily

$$Q_{pp} \;\ge\; \sum_j (c_j q_{pj}^2 + r_{pj}(2q_{pj} + 1)). \tag{4.27}$$

Indeed the smallest sum of squares of c_j nonnegative integers with sum T_{pj} occurs when their values differ by at most one. More strongly, subtracting the type averages shows that

$$Q - T\ \mathrm{diag}(c_j^{-1})\, T^{\mathsf{T}} \quad \text{is positive semidefinite.}$$

If this difference is zero, every b_{pW} is forced to equal T_{pj}/c_j within its restriction type. Integrality of these averages and both identities in (4.26) then give a complete exact test of this particular low-product completion. This is a necessary condition on center data; rejecting a single completion does not reject its input fusion ring.

For $R_{3,215}^{(6)}$ the released induction witness has six low objects and sixteen high objects. All its low character rows are real. Their central S rows are therefore real, and $S_{a^*,W} = \overline{S_{a,W}}$ together with invertibility of S forces these low objects to be self-dual in any realization. The automated fourth-order test produces a 15×15 matrix Q of rank 3. There are three independent high restriction types, with multiplicities $(4, 8, 4)$, and the type-variance matrix is exactly zero. Every forced average is a nonnegative integer. Expanding the three types into all sixteen formal objects gives an explicit integral matrix B satisfying both equations exactly.

The calculation and its frozen certificate are `research/m8_frontier/low_product_gram.py` and `research/m8_frontier/low_product_gram_certificate.json`. The program reconstructs the tensor, character table, inherited 230-equation witness, every cubic and fourth-order moment, all product restrictions, and the expanded Gram identities. Thus the stronger test survives and contributes zero general exclusions. It supplies no full center fusion tensor, individual twists, associators or categorification. The earliest general unresolved multiplicity remains three; further conditions or a categorical construction are still required there.

5 Module categories and Morita constraints

These criteria use module categories forced by algebra objects and complement the central-induction system. They do not impose a pivotal structure on the original category. Field-automorphism and tensor-sign reductions are kept with the arithmetic reductions in Section 3.

5.1 An involution-module obstruction

Theorem 5.1 (Antiinvariant module representation)**.** *Let R be a commutative fusion ring and let $g \ne 1$ be an invertible basis element with $g^2 = 1$. Suppose the set $J = \{x : gx = x\}$ is nonempty. If R has a complex fusion categorification, there are integer matrices $T_i \in M_{|J|}(\mathbb{Z})$, indexed by all basis elements, satisfying*

$$T_0 = I, \qquad T_i T_j = \sum_k N_{ij}^k T_k, \qquad T_{i^*} = T_i^{\mathsf{T}}, \tag{5.1}$$

$$|(T_i)_{xy}| \le N_{ix}^y, \qquad (T_i)_{xy} \equiv N_{ix}^y \pmod 2 \qquad (x, y \in J). \tag{5.2}$$

Failure of this finite system is a general categorification obstruction.

Proof. Let $\mathcal{C}$ be a hypothetical categorification. The additive category generated by a fixed simple x is a rank-one module category over the pointed subcategory generated by g. Choose isomorphisms $g \otimes x \simeq x$. The scalar module-associativity equation says that the pointed associator three-cocycle is a coboundary. Thus that subcategory is tensor equivalent to Vec_{C_2}, and $A = 1 \oplus g$ admits the regular connected separable algebra structure. This is a use of module associativity, not an assumption that g has a central lift. Standard separable-algebra module theory is as in [15, Chapter 7].

Let $\mathcal{M} = \mathcal{C}_A$ be the category of right modules. Its free and forgetful functors I_A, F_A satisfy $F_A I_A(X) = X \oplus (X \otimes g)$. Adjunction shows that a two-element g-orbit gives one simple module, whose restriction is the sum of the two simples in the orbit. A fixed simple x gives two nonisomorphic simples M_x^+, M_x^-, each restricting to x. Indeed, $\dim \mathrm{End}(I_A(x)) = 2$, so the squared multiplicities in its semisimple decomposition sum to two. The mixed Gram entries exclude constituents from other orbits. Every simple module occurs in a free module because A is separable.

The algebra automorphism α acting as $+1$ on 1 and -1 on g induces an involution Q of $\mathcal{M}$ by twisting the right action. It fixes the module attached to a two-element orbit and interchanges M_x^+ and M_x^-. The latter assertion follows directly because the nonzero action map $x \otimes g \to x$ changes sign, whereas an endomorphism of the simple underlying object is scalar. This twisting functor commutes with the left action of $\mathcal{C}$.

Consequently the (-1)-eigenspace of Q on $K_0(\mathcal{M}) \otimes \mathbb{Q}$ is a representation of R, with basis $e_x = [M_x^+] - [M_x^-]$, $x \in J$. Between two fixed pairs the action matrix has a block $\left(\begin{smallmatrix} a & b \\ b & a \end{smallmatrix}\right)$. Forgetting modules gives $a + b = N_{ix}^y$, while its action on the antiinvariant basis has coefficient $a - b$. Nonnegativity and integrality of a, b give (5.2). The module associator gives the multiplication identity in (5.1). Finally module Frobenius reciprocity makes the action of i^* the transpose of the action of i. All e_x have the same squared length two, so the restricted matrices obey the same transpose identity. No spherical or unitary structure has been used. □

Proposition 5.2 (Exact generator enumeration)**.** *The conditions of Theorem 5.1 can be decided by a finite integer search. If a selected set of basis elements generates $R \otimes \mathbb{Q}$ as an algebra, it suffices to enumerate their matrices, reconstruct the other matrices by rational polynomial expressions, and check* (5.1)–(5.2)*.*

Proof. Every entry has an explicit finite interval and parity. Choose monomials $w_0 = 1, w_1, \ldots, w_{r-1}$ forming a rational basis of the rank-r algebra, and let V have as its rows their coefficients in the based-ring basis. The expression for b_i is row i of V^{-1} applied to the monomials. Thus any representation is uniquely reconstructed from the selected generator matrices. Testing all the ring relations, transpose identities, and entry bounds after reconstruction accepts every possible solution and nothing outside the stated necessary system. □

The implementation treats $|J| \le 2$ and chooses one or two generators when such a set is found. It reports no conclusion otherwise. Applying it to the residual census gives nine complete exclusion certificates, including $R_{3,79}^{(6)}$ and $R_{3,216}^{(6)}$. The respective generator searches contain 4608 and 1536 assignments. These are automatic applications of one theorem; their indices are not inputs to the test.

5.2 Small-dimension Morita obstructions

The next criteria use categorical Morita equivalence rather than additional indicator-height searches. They apply to general complex fusion categories. In particular, no chosen pivotal structure enters their hypotheses.

5.2.1 A cyclic-index bound and dimension thirty

For group-theoretical data (G, ω, H, ψ), the simple objects of $\mathcal{C}(G, \omega, H, \psi)$ are indexed by a double coset HgH and an irreducible projective representation π of $S_g = H \cap gHg^{-1}$. Their Frobenius–Perron dimensions are

$$[H : S_g] \dim \pi. \tag{5.3}$$

The objects supported on H form a fusion subcategory equivalent to $\mathrm{Rep}(H)$. One can see this after replacing the data by an equivalent adapted-cocycle presentation with $\psi = 1$ and $\omega|_{G\times G\times H} = 1$; see [38, Proposition 2.5]. These descriptions also follow from [20].

Lemma 5.3 (Cyclic-index bound)**.** *Suppose a finite group G has a cyclic normal subgroup B of index at most two. Every simple object of every group-theoretical category $\mathcal{C}(G, \omega, H, \psi)$ has Frobenius–Perron dimension at most two.*

Proof. Put $K = H \cap B$. Since every subgroup of a cyclic group is characteristic, K is normal in G. Thus $K \subseteq S_g \subseteq H$ and $[H : K] \leq 2$. If $[H : S_g] = 2$, then $S_g = K$ is cyclic. Every irreducible projective representation of a cyclic group over $\mathbb{C}$ has dimension one, so (5.3) is two.

If $S_g = H$, it suffices to bound the dimensions of projective representations of H. In the twisted group algebra of H, the subalgebra on K is commutative and semisimple: its cyclic generator satisfies $u^{|K|} = c \neq 0$, a polynomial with distinct roots. The remaining coset, when present, permutes its weight spaces in orbits of size at most two. In an irreducible module there is one such orbit. For an orbit of size one, a coset generator squares to a nonzero scalar and hence has a one-dimensional invariant subspace. For an orbit of size two, a line in one weight space and its translate span an invariant two-dimensional subspace. Irreducibility therefore gives dimension at most two. This argument includes arbitrary projective factor sets. □

Proposition 5.4 (Integral dimension thirty)**.** *Every simple object of an integral fusion category of Frobenius–Perron dimension* 30 *has dimension one or two.*

Proof. The category is group-theoretical by [16, Theorem 9.2]. Its presenting group has order thirty. Every group of order thirty is solvable: the possible numbers of Sylow five- and three-subgroups are respectively $1, 6$ and $1, 10$; if neither is one, the corresponding nonidentity elements already number $24 + 20 > 29$. A normal Sylow subgroup and the solvable quotient prove the claim. Hall's theorem now supplies a subgroup of order fifteen. It is cyclic by the Sylow theorems and normal because it has index two. Lemma 5.3 applies. □

The adjective *integral* is essential: the statement does not claim the same bound for nonintegral categories of total dimension thirty. The certificate checks a positive integral dimension character, not just an integer value of the total dimension.

5.2.2 A Morita dichotomy in dimension sixty

Lemma 5.5. *A fusion category of Frobenius–Perron dimension sixty is either solvable or categorically Morita equivalent to $\mathrm{Vec}^{\omega}_{A_5}$ for some three-cocycle ω. If it is integral and solvable, has exactly two invertible simples, and satisfies $\mathcal{C}_{\mathrm{ad}} = \mathcal{C}$, every simple has dimension at most four.*

Proof. By [16, Theorem 9.16], the category is weakly group-theoretical. It is therefore Morita equivalent to a nilpotent category of the same dimension, obtained from vector spaces by successive extensions by finite groups whose orders multiply to sixty. If all these groups are solvable, refining their normal series into cyclic prime-order factors shows that the category is solvable. Solvability is invariant under categorical Morita equivalence; see [16, Section 4.2].

Otherwise an extension group has nonsolvable order dividing sixty. All proper divisors involve at most two primes, apart from thirty, which was handled above. Burnside's theorem and the fact that the unique nonabelian simple group of order sixty is A_5 imply that this group is A_5 itself. Its order uses the entire dimension. The nilpotent category is consequently pointed, of the form $\mathrm{Vec}^{\omega}_{A_5}$.

For the second assertion, a nontrivial last step of a solvable construction is a cyclic prime-order extension or equivariantization. An extension would give a nontrivial grading, contrary to $\mathcal{C}_{\mathrm{ad}} = \mathcal{C}$. In an equivariantization $\mathcal{C} = \mathcal{D}^{C_p}$, the canonical $\mathrm{Rep}(C_p)$ is a fusion subcategory. Exactly two invertible simples force $p = 2$, and $\mathrm{FPdim}(\mathcal{D}) = 30$.

The simples of a C_2-equivariantization are indexed by orbits of simples of $\mathcal{D}$ and projective representations of their stabilizers. Those stabilizers are trivial or cyclic of order two, so the projective representations have dimension one. A simple dimension in $\mathcal{C}$ is therefore either d or $2d$, with d a simple dimension in $\mathcal{D}$. Since $\mathcal{C}$ is integral, d is rational; as an algebraic integer it is an integer. Thus $\mathcal{D}$ is integral. Proposition 5.4 gives $d \leq 2$, proving the bound four. □

5.2.3 Cocycle-independent involution returns

For a normalized three-cocycle on a group of order two, its value on the triple of nonidentity elements is unchanged by a normalized coboundary. In particular, triviality of its restriction class implies that this value is one.

Theorem 5.6 (Involution-return indicator)**.** *Let $\mathcal{C}(G, \omega, H, \psi)$ be group-theoretical, let $g \in G$, and put $S = H \cap gHg^{-1}$. Suppose every element in*

$$T_g = \{r \in gH : r^2 \in S\}$$

is an involution conjugate in G to an element of H. For the simple object $M(g, \pi)$ associated to an irreducible projective representation π of S, one has

$$\nu_2(M(g, \pi)) = \frac{|T_g| \dim \pi}{|S|}. \tag{5.4}$$

In particular, if $T_g \neq \varnothing$, these simples are self-dual. This conclusion holds for every admissible ω, ψ.

Proof. Use the equivalent adapted presentation supplied by [38, Proposition 2.5]; its cocycle restricts to one on H^3. If r is an involution conjugate into H, the restriction class on $\langle r \rangle$ is trivial. The preceding coboundary observation gives $\omega(r, r, r) = 1$ in this adapted presentation, independently of any remaining values of the cocycle.

The second-indicator specialization of [38, Theorem 4.2, equation (4.6)] is

$$\nu_2(M(g, \pi)) = \frac{1}{|S|} \sum_{r \in T_g} \pi_{-2}(r) \chi_\pi(r^{-2}).$$

For an involution, the scalar $\pi_{-2}(r)$ in that formula is $\omega(r, r, r)^{-1} = 1$, whereas $\chi_\pi(r^{-2}) = \chi_\pi(1) = \dim \pi$. This proves (5.4). Group-theoretical categories have their positive spherical structures; a simple with nonzero second indicator is self-dual. Self-duality is preserved under tensor equivalence, so the conclusion is independent of the presentation. □

Corollary 5.7 (Dihedral supports in A_5)**.** *In every category $\mathcal{C}(A_5, \omega, D_{2q}, \psi)$ with $q \in \{3, 5\}$, every simple of dimension q has second indicator one. In particular all such simples are self-dual.*

Proof. The dihedral subgroup has order $2q$. Its ordinary irreducible representations have dimensions one or two, so a simple of dimension $q > 2$ is not supported on it. Formula (5.3) then forces stabilizer order two and a one-dimensional projective representation. For these double cosets the following exact group data hold:

$\lvert H\rvert$	double-coset sizes	$\lvert S\rvert$	$\lvert T_g\rvert$ when $\lvert S\rvert = 2$
6	$6, 18, 36$	2	2
10	$10, 50$	2	2

Every element in T_g is an involution. All fifteen involutions in A_5 are conjugate, and H contains an involution. Thus Theorem 5.6 gives $\nu_2 = 2/2 = 1$.

These group assertions are certified independently from the sixty even permutations of five letters. The verifier generates every order-$2q$ subgroup using an element of order q and an involution, checks that they form one conjugacy class, and enumerates every right coset and every square-return element. It does not test only a selected coset or an assumed cocycle. The coset lists are included in each certificate. □

Theorem 5.8 (Tensor-checkable small-dimension obstruction)**.** *Let R be an integral fusion ring. Each of the following conditions excludes a general complex categorification.*

(a) $\mathrm{FPdim}(R) = 30$ *and some basis dimension exceeds two.*

(b) $\mathrm{FPdim}(R) = 60$*, the adjoint based subring is R, and exactly two basis elements are invertible. Some basis dimension exceeds four. For a prime $q \in \{3, 5\}$ the complete set of based-subring dimensions is $\{1, 2, 2q, 60\}$, and a basis element of dimension q is not self-dual.*

Proof. Part (a) is Proposition 5.4. For (b), Lemma 5.5 excludes solvability and gives a presentation $\mathcal{C}(A_5, \omega, H, \psi)$. The canonical $\mathrm{Rep}(H)$ subcategory forces $|H| \in \{1, 2, 2q, 60\}$. If $|H| \leq 2$, every simple dimension is at most two, contradicting the stated dimension exceeding four. If $H = A_5$, the category is $\mathrm{Rep}(A_5)$ and has only one invertible simple, because A_5 is perfect. Thus H is an order-$2q$ subgroup, necessarily a conjugate of D_{2q}. Corollary 5.7 contradicts the non-self-dual dimension-q basis element. □

5.2.4 Uniform census application

The implementation supplies the same tensor test at every multiplicity. It first checks a positive integral dimension character; it then computes the adjoint closure, all based subrings, duality, and the number of invertibles. Illustrative applications are

ring	sorted dimension type	branch	q
$R^{(6)}_{2,51}$	$(1, 1, 2, 2, 5, 5)$	(b)	5
$R^{(6)}_{3,15}$	$(1, 1, 2, 3, 3, 6)$	(b)	3
$R^{(6)}_{4,19}$	$(1, 1, 1, 1, 1, 5)$	(a)	–

The program finds these applications from the tensor, without selecting labels in advance. The current census scan and its exact dimension checks are recorded by `software/morita_controls.py`. The criterion also excludes $R^{(6)}_{2,25}$ and $R^{(6)}_{2,29}$. Overlap with other criteria is counted only once in the sequential exclusion table.

5.3 Two-sided module Gram constraints

Let $\mathcal{C}$ be a fusion category, let A be a connected separable algebra, and let $\mathcal{M} = \mathcal{C}_A$. Write M_i for the action matrix of a simple $X_i \in \mathcal{C}$, and $\mathcal{D} = \mathrm{Fun}_{\mathcal{C}}(\mathcal{M}, \mathcal{M})$ for the dual category, with a

fixed choice of opposite tensor convention. All dimension statements here use Frobenius–Perron dimensions, not a positive pivotal structure.

Proposition 5.9 (Two-sided module Gram identities)**.** *For every simple $m \in \mathcal{M}$, let B_m be the matrix of the evaluation functor $\mathcal{D} \to \mathcal{M}$, $Y \mapsto Y(m)$. Then*

$$B_m B_m^{\mathsf{T}} = \sum_i (M_i)_{mm} M_i. \tag{5.5}$$

The tensor unit column of B_m is e_m. If d_n are the positive module dimensions in any fixed normalization, each column v belonging to a simple $Y \in \mathcal{D}$ satisfies

$$\sum_n v_n d_n = d_m \operatorname{FPdim}(Y). \tag{5.6}$$

Proof. The internal endomorphism algebra of m in $\mathcal{C}$ is $A_m = \bigoplus_i (M_i)_{mm} X_i$, up to the harmless replacement of every index by its dual; these multiplicities agree by module reciprocity. The module category is equivalently $\mathcal{C}_{A_m}$, sending m to the free rank-one module. The dual category is the category of A_m-bimodules. Evaluation becomes the functor forgetting the left action, whose adjoint is free left multiplication by A_m. Their composite on $\mathcal{M}$ is therefore the action of A_m. Adjunction gives (5.5). Evaluation of the tensor unit is m, giving the specified column. The commuting action of the dual category on the positive module dimension vector has eigenvalue $\operatorname{FPdim}(Y)$; this gives (5.6). These assertions are standard consequences of separable-algebra Morita theory; see [15, Chapter 7]. □

5.3.1 The split-involution input and a finite system

Suppose $g^2 = 1$ is an invertible object fixing a simple object of $\mathcal{C}$. The module-associativity argument in the main manuscript supplies the algebra $A = 1 \oplus g$. Free two-element g-orbits give one simple of $\mathcal{M}$; fixed simples give two. The sign automorphism of A interchanges the two actions on every fixed simple. Thus the action matrices M_i can be reconstructed from the full tensor and an anti-invariant representation satisfying the parity, transpose and entry bounds of Theorem 5.1.

Normalize the dimension of the free module A to one. An orbit module has dimension $\operatorname{FPdim}(X)$ and a split fixed module has dimension $\operatorname{FPdim}(X)/2$. These module dimensions need not be algebraic integers. At $m = A$, the Gram matrix in (5.5) is $I + M_g$. Since M_g is an involutive permutation, its nonnegative integral Gram columns are forced: a fixed point has two unit columns, and a transposed pair has one column supported on that pair. Hence (5.6) determines the multiset of simple dimensions of $\mathcal{D}$. One of the unit-dimension columns is its tensor unit.

For another simple m, remove its known tensor-unit column. For every remaining prescribed dual dimension e, enumerate nonzero integer vectors $v \geq 0$ with

$$v_i^2 \leq Q_{ii}, \quad v_i v_j \leq Q_{ij}, \quad \sum_i v_i d_i = e d_m, \qquad Q = \sum_i (M_i)_{mm} M_i - e_m e_m^{\mathsf{T}}.$$

The integer multiplicities of these columns must add their outer products to Q and respect the prescribed number of columns of each dual dimension. Relaxing the multiplicities to nonnegative rationals gives a necessary rational linear system. An exact Farkas vector proves infeasibility. A general exclusion requires infeasibility for every admissible anti-invariant representation; selecting only one is invalid.

The executable implementation treats at most two fixed simples and simple-dimension fields of degree at most four. It enumerates all admissible generator matrices and reconstructs the complete representation. The vector search uses exact field coordinates and explicit integer

bounds. A domain-budget limit or an ineligible fixed-point count gives no conclusion. Negative checking, when a certificate is proposed, reconstructs the full domain and checks only exact arithmetic.

The implementation in `software/module_dual_gram.py` is a uniform necessary-condition test. It does not assign categorical existence when the relaxed system is feasible. Its successful certificates, if any, must enter the same exact-checking and disjoint allocation process as every other obstruction.

6 Categorical models and exact tensor reconstruction

Positive records are established by tensor matches to constructions, not by lack of an obstruction. We use Deligne products of the standard rank-two and rank-three unitary categories, the TT_3 category of [13, Definition 2.5], and the positive quadratic constructions of [28, Proposition 5.1, Corollary 5.8]. We use the latter results only for existence, not as a negative classification oracle. Every displayed tensor is reconstructed from its defining rules and compared under unit-preserving permutations with the census.

6.1 A central fermion and a cyclic-four quadratic ring

Let $\mathcal{D}$ be the unitary near-group category $\mathrm{Ad}(E_6)$ with basis $1, h, X$ and rules

$$h^2 = 1, \qquad hX = Xh = X, \qquad X^2 = 1 + h + 2X. \tag{6.1}$$

Its noninvertible dimension is $d = 1 + \sqrt{3}$, and $D = \dim \mathcal{D} = 4 + 2d$.

Lemma 6.1 (Central fermion)**.** *There is an invertible $z \in Z(\mathcal{D})$ with $F(z) = h$, $z^2 = 1$, and $\theta_z = -1$. The full subcategory generated by z is the unitary symmetric category* sVec.

Proof. The formal codegrees are D, $D' = 6 - 2\sqrt{3}$, and 2. Thus $I(1) = 1 \oplus A \oplus B$, with dimensions $1, 1 + d, 2 + d$. Since $FI(1) = 3 \cdot 1 + h + 2X$, comparison of the unit multiplicities and of the coefficients of $1, d$ gives

$$F(A) = 1 + X, \qquad F(B) = 1 + h + X.$$

All three low twists are one.

The induction Gram matrix in the basis $(1, h, X)$ is

$$\begin{pmatrix} 3 & 1 & 2 \\ 1 & 3 & 2 \\ 2 & 2 & 8 \end{pmatrix}.$$

Only B among the low objects occurs in $I(h)$, once. The remaining endomorphism dimension is $3 - 1 = 2$, so the remainder consists of two distinct simples, both with multiplicity one. Its restriction is $FI(h) - F(B) = 2h + X$. Consequently these two objects are z, W with $F(z) = h$ and $F(W) = h + X$. The object z has dimension one and is invertible. Its square restricts to the unit; the only central simple restricting to the unit is the unit itself, as the decomposition of $I(1)$ shows. Therefore $z^2 = 1$.

The pointed associator class on $\langle h \rangle$ is trivial: the rank-one module category on X supplies a two-cochain whose coboundary is the restricted associator. In the positive spherical structure, $\nu_2(h) = 1$. Equation (4.6) at $k = 2$ implies that every twist in $I(h)$ squares to one, because all dimensions are positive and their total is D. At $k = 1$ it gives

$$0 = (2 + d) + \theta_z + (1 + d)\theta_W.$$

The known positive summand has exactly the total dimension of the other two. Equality in the triangle inequality forces $\theta_z = \theta_W = -1$. Finally an order-two invertible in a unitary ribbon category with twist -1 has self-braiding -1 in its canonical positive structure. Its generated pointed braided category is sVec. □

Proposition 6.2 (Fermionic zesting)**.** *The ring $R^{(6)}_{2,2}$ has a unitary categorification. More generally, if a unitary fusion category $\mathcal{D}$ has a central fermion z of order two, the rules*

$$(U, a) \star (V, b) = (U \otimes V \otimes F(z)^{ab}, a + b), \qquad a, b \in \mathbb{Z}/2, \tag{6.2}$$

on $\mathcal{D} \boxtimes \mathrm{Vec}_{\mathbb{Z}/2}$ admit a unitary associator.

Proof. Use the central two-cocycle $\lambda(a, b) = z^{ab}$ and normalized associativity scalar $\tau(a, b, c) = i^{abc}$, where the exponents use representatives $0, 1$. The only central interchange obstruction is $c_{\lambda(a,b),\lambda(c,d)} = (-1)^{abcd}$. A direct calculation gives

$$\begin{aligned} bcd + a(b \oplus c)d + abc - (a \oplus b)cd - ab(c \oplus d) \\ \equiv 2abcd \pmod 4, \end{aligned}$$

where $\oplus$ is addition modulo two. Thus the coboundary of τ is precisely that interchange obstruction. The associative-zesting construction of [9, Definition 3.2, Proposition 3.4], in the relative-center formulation also recalled in [3, Definition 2.1], therefore supplies the pentagon and rigidity for (6.2). All structural maps used are unitary, including the scalar τ, so the resulting fusion category is unitary. This use of a known construction does not claim to have independently solved its individual F-symbol equations.

Apply Lemma 6.1. Let $g = (1, 1)$, identify $g^2 = (h, 0)$, and write $Y = (X, 0)$ and $X_1 = (X, 1)$. Then the invertibles form C_4,

$$gY = X_1, \quad gX_1 = Y, \quad Y^2 = X_1^2 = 1 + g^2 + 2Y, \quad X_1 Y = g + g^3 + 2X_1.$$

These rules reconstruct the full tensor that is based-isomorphic to $R^{(6)}_{2,2}$. The supplement checks that equality and all sixteen scalar coboundary identities exactly. □

The proposition is a positive construction, not an assertion that this fusion category or its ring has not appeared in another presentation. It is particularly useful here because a ring that survived the first obstruction pass is thereby removed from the residual list for the correct reason: existence rather than an attempted exclusion.

6.2 Finite-group and group-theoretical models

For $G = (C_3 \times C_3) \rtimes C_4$, with the order-four rotation $(x, y) \mapsto (-y, x)$, the character ring is reconstructed from four linear characters and two induced characters. Exact character inner products give all fusion coefficients and identify $R^{(6)}_{2,11}$.

We also compute the untwisted group-theoretical categories $\mathcal{C}(G, 1, H, 1)$ for

$$(G, H) = (A_4, C_2), \quad (A_5, D_{10}), \quad (A_5, S_3).$$

Here D_{10} has order ten. The finite-group construction is standard; we recall its explicit tensor calculation to distinguish it from a mere comparison of dimension types. Realize the category as H-bimodules in Vec_G. Its simples are supported on double cosets HgH and indexed by irreducible representations of $H \cap gHg^{-1}$. The tensor product is relative tensor product over the group algebra of H. Equivalently it is convolution of the corresponding equivariant vector bundles. The character of a convolution fiber is computed by summing over the middle

H-orbits and taking invariants under the stabilizer; averaging the finite stabilizer action gives its multiplicities. This is a finite exact character-inner-product calculation. The code lists the groups as permutations, computes the double cosets and stabilizers, and uses their exact character values. It then checks the resulting complete fusion tensors.

The three outputs are respectively $R^{(6)}_{2,36}$, $R^{(6)}_{2,105}$, and $R^{(6)}_{3,141}$. The last model is also discussed in [2, Section 6.5]. Finite groups, ordinary unitary representations, and relative tensor product over a finite-dimensional C^*-algebra give unitary fusion categories in these untwisted cases.

6.3 ADE modules, cyclic Izumi models, and a finite group

The following constructions add five realizations at multiplicity two and one at multiplicity three. Each constructor builds its fusion tensor without reading the census, then searches for a full based-ring match.

Proposition 6.3 (ADE modules and their zesting)**.** *The rings $R^{(6)}_{2,75}$, $R^{(6)}_{2,14}$, $R^{(6)}_{2,128}$ and $R^{(6)}_{2,140}$ admit unitary categorifications. They are, respectively, the E_6 module category over $SU(2)_{10}$, its fermionic grading zest, the D_6 module category over $SU(2)_8$, and the local D_{10} module category over $SU(2)_{16}$.*

Proof. Use the unitary $SU(2)_k$ category with simples $V_0, \dots, V_k$ and fusion coefficients

$$N_{ij}^{l} = 1 \quad \Longleftrightarrow \quad |i-j| \le l \le \min(i+j, 2k-i-j), \quad i+j+l \equiv 0 \pmod 2.$$

The E_6 algebra is $V_0 \oplus V_6$ at level ten. The D-type algebra is $V_0 \oplus V_k$ at levels divisible by four. Their existence, separability, module categories and the ADE module graphs are the Kirillov–Ostrik construction [26, Theorem 6.1 and Section 7]. These algebras and their module tensor products have the standard unitary structures.

For E_6, label the graph vertices $0, \dots, 5$ and its edges

$$01, \quad 12, \quad 23, \quad 34, \quad 25,$$

with unit vertex zero. Multiplication by the free image of V_1 is the adjacency matrix B. The rows $(B^j)_{0,*}$, $0 \le j \le 5$, form an invertible matrix V. Centrality of the free image of V_1 and the cyclicity of the unit vector imply that all multiplication matrices are uniquely determined by $N_i = \sum_j (V^{-1})_{ij} B^j$. The exact constructor evaluates these matrices, checks nonnegative integral entries and all tensor axioms, and matches $R^{(6)}_{2,75}$.

The parity of the original $SU(2)$ labels descends to the module category; in these coordinates it is $(0,1,0,1,0,1)$. The free image of V_{10} is the degree-zero invertible at vertex four. The canonical central free functor sends the order-two fermion V_{10} to a central fermion g in the module category. Hence associative zesting may multiply the product of two odd objects by g. Its obstruction is the fermionic sign $(-1)^{abcd}$, trivialized by the scalar cochain i^{abc}, exactly as in Proposition 6.2. All sixteen cochain identities are checked. This constructs a unitary fusion category and its tensor is $R^{(6)}_{2,14}$. The construction uses the actual grading and central fermion, not just a ring automorphism.

For the D cases put $m = k/2$. The orbit modules are O_j, $0 \le j < m$, and the fixed object V_m splits as $U_+ \oplus U_-$. The free image of V_j is $O_{\min(j,k-j)}$ unless $j = m$. Products of two orbit modules are obtained by applying this induction rule to the displayed $SU(2)$ coefficients. For an orbit module times $U_+ \oplus U_-$ use the same rule; the difference $U_+ - U_-$ has eigenvalue zero for odd j and $(-1)^{j/2}$ for even j. At the two levels used here, $k \equiv 0 \pmod 8$, the resolved objects are self-dual. The fixed-point resolution formulas of [26, Section 7, D_{2m} fusion table] give the remaining products. Equivalently, in the local sector of this congruence class,

$$U_\pm^2 = U_\pm + O_0 + O_4 + \cdots + O_{m-4}, \qquad U_+U_- = O_2 + O_6 + \cdots + O_{m-2}.$$

Together with Frobenius reciprocity these formulas are the constructor's complete specification. At $k = 8$ use all six modules; at $k = 16$ use the local modules $O_0, O_2, O_4, O_6, U_+, U_-$. Their tensors match $R^{(6)}_{2,128}$ and $R^{(6)}_{2,140}$ respectively. □

Proposition 6.4 (The Izumi cyclic-five model)**.** *The ring $R^{(6)}_{2,143}$ is the Grothendieck ring of the commutative even part of the unitary Izumi C_5 subfactor.*

Proof. The existence of the cyclic-five Izumi subfactor and the relevant even part are established in the literature summarized in [21, Section 6.6]. Its commutative fusion rules are those in [17, Theorem 7]. To specify the full tensor used for the match, write $1, a, b_1, b_2, c_1, c_2$, identify indices i and $-i$ modulo five, and put $b_0 = c_0 = 0$, $\Sigma = a + b_1 + b_2 + c_1 + c_2$. The products are

$$\begin{gathered} a^2 = 1 + \Sigma, \qquad ab_i = \Sigma - b_i, \qquad ac_i = \Sigma + c_i, \qquad b_i c_j = \Sigma, \\ b_i b_j = \delta_{ij} 1 + (1 - \delta_{ij}) a + \Sigma - a - b_{i+j} - b_{i-j}, \\ c_i c_j = \delta_{ij} 1 + (1 + \delta_{ij}) a + \Sigma - a + c_{i+j} + c_{i-j}. \end{gathered}$$

All products commute and all simples are self-dual. The independent constructor specializes these formulas at five and verifies all 216 coefficients of the matched tensor. Existence is the cited subfactor construction, not a consequence of the based-ring axioms alone. □

Proposition 6.5 (An exact finite-group model)**.** *The ring $R^{(6)}_{3,47}$ is categorified by* $\mathrm{Rep}(GL(3, \mathbb{F}_2)) \simeq \mathrm{Rep}(PSL(2, 7))$.

Proof. Generate the group as all invertible 3×3 binary matrices, with actual matrix multiplication. It has 168 elements and six conjugacy classes of sizes $1, 21, 56, 42, 24, 24$, ordered by element orders $1, 2, 3, 4, 7, 7$. Put $a = (-1 + \sqrt{-7})/2$ and $b = \overline{a}$. The candidate character table is

	1	2	3	4	7	7
1	1	1	1	1	1	1
3	3	−1	0	1	a	b
$\overline{3}$	3	−1	0	1	b	a
6	6	2	0	0	−1	−1
7	7	−1	1	−1	0	0
8	8	0	−1	0	1	1

.

We verify its representation-theoretic meaning, not just orthogonality. The program computes the multiplication constants of the six class sums in $\mathbb{C}[GL(3, \mathbb{F}_2)]$. For each row χ it checks that $|C|\chi(C)/\chi(1)$ is a unital character of this class algebra. The six characters are distinct and therefore exhaust its primitive central characters. By semisimplicity they are the normalized central characters of the actual irreducible representations. Row orthogonality and positivity of the displayed degrees fix the normalization, so the rows are the actual irreducible characters. Taking their product inner products gives nonnegative integer structure constants, evaluated exactly in $\mathbb{Q}(\sqrt{-7})$. The resulting full tensor is based-isomorphic to $R^{(6)}_{3,47}$. A finite-group representation category has its usual unitary realization. □

6.4 Categorical structures, extensions, and central quotients

The following determinations use categorical constructions, followed by exact identification of their full fusion tensors. A based-ring isomorphism transports both existence and a proved obstruction to a specified categorical structure. It does not turn spherical existence into pseudo-unitary existence. All identifications in this section are recomputed by `software/literature_models.py`; no census label is an input to the construction.

6.4.1 Schopieray's spherical model

Schopieray constructs the rank-six modular category $\mathcal{C}(\mathfrak{so}_5, 3/2)_{\mathrm{ad}}$ in [41, Example 1, equations (1)–(2)]. Its displayed fusion matrices, in the order $(X_0, \dots, X_5)$, give our tensor $R^{(6)}_{2,124}$ after the permutation

$$(0, 1, 5, 4, 2, 3).$$

Here, and in the electronic catalogue, a permutation p means $N_{ij}^{\mathrm{census}\,k} = N_{p(i),p(j)}^{\mathrm{source}\ p(k)}$. There can be more than one such permutation; the catalogue stores the lexicographically first one. The six matrices are independently transcribed in the constructor and all 216 coefficients are compared. Schopieray's Theorem 1 proves that this based ring admits no pseudo-unitary categorification. Its exact existence record is therefore

	general	pivotal	spherical	pseudo-unitary	unitary
$R^{(6)}_{2,124}$	yes	yes	yes	no	no.

The last entry follows because a unitary fusion category is pseudo-unitary. This is a ring-level nonexistence theorem, not only the failure of a particular spherical structure to have positive dimensions. It does not contribute to the count of *general* exclusions.

6.4.2 A nilpotent extension and an Izumi even part

Write H_8 for the Kac–Paljutkin Hopf algebra of dimension eight. Choi–Lu–Sun give extensions of $\mathrm{Rep}(H_8)$ with one additional simple object in [7, equation (1.4), Section 7.1 and Tables 3–4]. With basis $(1, a, b, ab, N, D)$ and $a^2 = b^2 = 1$, their fusion rules are

$$\begin{aligned} gN = N, \quad gD = D, \quad N^2 &= \sum_{g \in C_2^2} g, \\ ND = DN = 2D, \qquad D^2 &= \sum_{g \in C_2^2} g + 2N. \end{aligned} \tag{6.3}$$

The relation $gN = N$ is meant for each $g \in C_2^2$, as is $gD = D$. The source provides existence; it is not inferred from associativity of these integer rules.

For a separate categorical justification of unitarizability, the successive adjoint subcategories determined by (6.3) have bases

$$\{1, a, b, ab, N, D\} \supset \{1, a, b, ab, N\} \supset \{1, a, b, ab\} \supset \{1\}.$$

Thus every categorification with these rules is nilpotent. A nilpotent fusion category is weakly group-theoretical and is unitarizable by [18, Theorem 5.20]. The program recomputes this adjoint series by support closure. The source tensor matches $R^{(6)}_{2,69}$ with the identity permutation. This proves unitary *existence*; neither nilpotency alone nor a tensor-axiom check is used to assert existence.

The dual even part of the Izumi C_4 subfactor is a unitary fusion category. Its multiplication table in [24, p. 27], in the source order $(1, \kappa, \beta\kappa, \chi, \sigma, \beta)$, matches $R^{(6)}_{2,98}$ after the permutation

$$(0, 5, 1, 2, 4, 3).$$

The constructor recovers every entry of that table before making the comparison. The equality is with $R^{(6)}_{2,98}$, not with the dimension-matching tensor $R^{(6)}_{2,41}$. Equality of dimension lists would not establish this identification.

6.4.3 A free central-boson quotient

A *central boson* here means an invertible object $\widetilde{g}$ in the unitary Drinfeld center $Z(\mathcal{D})$, generating a symmetric subcategory equivalent to $\mathrm{Rep}(C_2)$, whose underlying object g in $\mathcal{D}$ is nontrivial. This includes the half-braiding and the Tannakian structure. Centrality of the basis element $[g]$ in $K_0(\mathcal{D})$ is not a substitute for this hypothesis.

Proposition 6.6 (Free central-boson quotient)**.** *Let $\mathcal{D}$ be a unitary fusion category and let $\widetilde{g}$ be a central boson as above. Suppose tensoring with g has no fixed point on* $\mathrm{Irr}(\mathcal{D})$*. Then the right modules over the central algebra $A = 1 \oplus g$ form a unitary fusion category $\mathcal{D}_A$. Its simples are indexed by the two-element g-orbits. For orbit representatives i, j, k its fusion coefficients are*

$$\widetilde{N}_{[i],[j]}^{[k]} = N_{ij}^{k} + N_{ij}^{gk}. \tag{6.4}$$

Proof. The algebra A is the image of the regular function algebra of C_2 under the central Tannakian embedding. It is connected and separable, and is commutative as a central algebra. The half-braiding makes right A-modules into a tensor category with product $\otimes_A$; see [15, Theorem 8.23.3]. Use the standard dagger Frobenius structure on the regular algebra. Work first with its unitary modules. Every algebraic module is a summand of a free module by separability; an idempotent in the finite-dimensional C^*-endomorphism algebra of a free unitary module is conjugate to an orthogonal projection. Thus these unitary modules represent the whole algebraic module category up to equivalence. Relative tensor products are the images of the self-adjoint separability projections. The unitary associativity maps of $\mathcal{D}$ descend to unitary maps between their images, and the module morphism spaces inherit the adjoint operation. Thus $\mathcal{D}_A$ is a unitary fusion category.

The free-module functor $F(i) = i \otimes A$ is a tensor functor. By adjunction,

$$\mathrm{Hom}_A(F(i), F(j)) \cong \mathrm{Hom}_{\mathcal{D}}(i, j \oplus jg).$$

For simple i, freeness of the g-action makes the endomorphism algebra of $F(i)$ one-dimensional, so $F(i)$ is simple. The same formula shows $F(i) \cong F(j)$ precisely when $j = i$ or $j = gi$. Every simple module is a summand of a free module, since A is separable, and hence is one of these simple free modules. Finally

$$F(i) \otimes_A F(j) \cong F(i \otimes j).$$

Collecting the two constituents k, gk in each orbit gives (6.4). It also proves independence of the chosen representatives. □

Apply the proposition to

$$\mathcal{D} = \mathrm{Ising} \boxtimes PSU(2)_6, \qquad g = (\psi, V_6).$$

In the principal unitary braidings, ψ and V_6 are order-two fermions. Their product has self-braiding $+1$ and generates a Tannakian $\mathrm{Rep}(C_2)$, with central lift supplied by the product braiding. The simple labels of $PSU(2)_6$ are V_0, V_2, V_4, V_6; tensoring by V_6 exchanges V_0, V_6 and V_2, V_4. Therefore the diagonal action on the twelve simples of $\mathcal{D}$ is free. Formula (6.4) gives a rank-six unitary category, whose tensor is $R_{2,94}^{(6)}$. The checker reconstructs the parent tensor, checks all twelve action labels, computes the six orbits and verifies the displayed formula and its independence of representatives. Its orbit order is specified in `data/diagonal_fermion_quotient.json`; the final census permutation is $(0, 3, 1, 2, 4, 5)$.

The construction uses *all* right A-modules with the tensor product supplied by the central lift. It does not identify them with the local modules of a braided algebra and does not assert that the quotient inherits the ambient braiding. In particular, no transparency hypothesis has been silently added or inferred from the fusion ring.

6.5 A diagonal fermion with a fixed point

The free-action formula of Section 6.4 does not apply to a fixed simple object. Its two resolutions can have different duality depending on the local rotation data. We give a calculation which resolves the fixed point and determines every fusion coefficient.

We use positive spherical structures in unitary categories. If $f^2 = 1$, $x^* = x$, and $t : f \otimes x \longrightarrow x$ is an isomorphism, denote by $t^{\mathsf{T}} : x \otimes f \longrightarrow x$ its pivotal transpose, using the specified self-dual identifications of f and x. Equivalently it is obtained from t by inserting a coevaluation for x on the right and evaluating the left pair. Associators are suppressed in this notation. Normalize t so that the action of two copies of f is the identity.

Lemma 6.7 (Duality of a resolved fixed point)**.** *Let (f, c) be a central boson of order two, and let V be a self-dual simple fixed by f. Suppose $t : fV \to V$ is a normalized action and*

$$t^{\mathsf{T}} c_{f,V} = \delta t, \qquad \delta \in \{1, -1\}. \tag{6.5}$$

For the central algebra $A = 1 \oplus f$, the two simple left A-modules with underlying object V have actions t and $-t$. Their duality is $V_\epsilon^* = V_{\delta\epsilon}$, where $\epsilon = \pm 1$.

Proof. The two actions satisfy the module equation because t is normalized. They are simple and inequivalent since V is simple and a scalar intertwiner cannot change the sign of its nonzero action. Their sum is the free module on V: the commuting algebra of that free module is the two-dimensional semisimple algebra of its stabilizer.

The dual left action is obtained by transposing the given action and then moving f from the left to the right of V^* with its specified half-braiding. This is exactly the composite in (6.5). The normalization $f^2 = 1$ makes the inverse action equal to the action, and the symmetric Frobenius structure of A supplies the ordinary pivotal transpose. Thus its sign is $\delta\epsilon$. Alternatively, on the one-dimensional ordinary vacuum channel of $V \otimes V$, the projector for the relative tensor product of signs ϵ, η has scalar $(1 + \epsilon\eta\delta)/2$. The unit occurs precisely when $\eta = \delta\epsilon$, giving the same duality rule. □

Theorem 6.8 (Equal-phase fermionic product)**.** *Let $\mathcal{E}_m$ and $\mathcal{E}_n$ be unitary fusion categories with respective self-dual bases $1, f, x$ and $1, f', y$ and rules*

$$f^2 = 1, \quad fx = xf = x, \quad x^2 = 1 + f + mx, \qquad (f')^2 = 1, \quad f'y = yf' = y, \quad y^2 = 1 + f' + ny.$$

Assume specified central fermions f, f' and normalized actions t, s satisfy

$$t^{\mathsf{T}} c_{f,x} = -it, \qquad s^{\mathsf{T}} c_{f',y} = -is. \tag{6.6}$$

The categories have specified positive pivotal identifications used in these equations. Then m, n are even, and the category of modules over the central algebra

$$A = 1 \oplus (f \boxtimes f') \quad \text{in} \quad \mathcal{E}_m \boxtimes \mathcal{E}_n$$

is a unitary fusion category of rank six. With basis $1, g, a, b, u, v$, its duality fixes $1, g, a, b$ and interchanges u, v. Its complete multiplication is determined by commutativity and

$$g^2 = 1, \quad ga = a, \quad gb = b, \quad gu = v, \quad gv = u, \tag{6.7}$$

$$a^2 = 1 + g + ma, \quad b^2 = 1 + g + nb, \quad ab = u + v, \tag{6.8}$$

$$au = av = b + \tfrac{m}{2}(u + v), \qquad bu = bv = a + \tfrac{n}{2}(u + v), \tag{6.9}$$

$$\begin{aligned} u^2 = v^2 &= g + \tfrac{m}{2}a + \tfrac{n}{2}b + \tfrac{mn}{4}(u + v), \\ uv &= 1 + \tfrac{m}{2}a + \tfrac{n}{2}b + \tfrac{mn}{4}(u + v). \end{aligned} \tag{6.10}$$

If $d_m = \mathrm{FPdim}(x)$ and $d_n = \mathrm{FPdim}(y)$, its dimensions are $1, 1, d_m, d_n, d_m d_n/2, d_m d_n/2$.

Proof. The product of the specified fermions is a central boson: its self-braiding is $(-1)(-1) = 1$. Its central regular algebra A is connected and separable. Modules over this algebra have relative tensor product, with tensor unit A, and form a unitary fusion category. This uses all modules over a *central* algebra; no braiding on the quotient is asserted. Write Q for the tensor free-module functor.

There are four free orbits and one fixed point in the nine simples of the Deligne product. The corresponding modules are

$$1 = Q(1,1), \quad g = Q(f,1), \quad a = Q(x,1), \quad b = Q(1,y), \qquad Q(x,y) = u \oplus v.$$

On the fixed point $V = x \boxtimes y$, use the action $t \boxtimes s$. Its square is one. The transpose and half-braiding in a Deligne product factor, so (6.6) gives

$$(t \boxtimes s)^{\mathsf{T}} c_{f\boxtimes f',V} = (-i)^2 (t \boxtimes s) = -(t \boxtimes s).$$

Lemma 6.7 proves $u^* = v$. The other four objects are self-dual because they are tensor free modules on self-dual objects. The module dimension equals the underlying dimension divided by $\dim A = 2$, giving the asserted dimensions and rank.

Tensoring a resolved fixed point by g changes its action sign. Indeed, to form the relative product with $Q(f,1)$ the diagonal algebra line is moved past $(f,1)$; the scalar is $c_{f,f} = -1$. Identifying $(f,1)V$ with V by $t \boxtimes \mathrm{id}$ leaves no other scalar, by the normalized square of t. Hence $gu = v$, $gv = u$. Taking duals gives $ug = v$, $vg = u$, so g is central on these objects. The free-module tensor identity gives $g^2 = 1$, $ga = ag = a$, $gb = bg = b$ and (6.8).

Since g fixes a, the two products au, av are equal, and likewise $ua = va$. Their respective sums, computed by the tensor functor Q, both equal $2b + m(u+v)$. This proves the first half of (6.9) on both sides and also forces m even. The second half and the evenness of n follow similarly. Thus all products involving a, b, g commute. Since $v = gu$ and g is central, u and v commute as well.

Now $u^2 = v^2$ and $(u^2)^* = v^2 = u^2$. The coefficient of 1 in u^2 is zero and that of g is one. Frobenius reciprocity and (6.9) give coefficients $m/2, n/2$ on a, b. Self-duality of u^2 makes the coefficients of u, v equal. Put $e = d_m d_n/2$ and call the common coefficient k. From $d_m^2 = 2 + m d_m$ and $d_n^2 = 2 + n d_n$ one obtains

$$e^2 = 1 + \tfrac{m}{2} d_m + \tfrac{n}{2} d_n + \tfrac{mn}{2} e.$$

Dimension comparison gives $k = mn/4$ and proves the square formula. Multiplication by g then gives the formula for uv. □

Corollary 6.9 (Three tensor identifications)**.** *The cases $(m,n) = (0,0), (0,2), (2,2)$ of Theorem 6.8 have unitary realizations and identify, respectively, the census rings*

$$R^{(6)}_{1,3}, \qquad R^{(6)}_{2,14}, \qquad R^{(6)}_{2,40}.$$

In particular, $R^{(6)}_{2,40}$ is the diagonal-fermion quotient of two identical *unitary half-E_6 categories.*

Proof. For $m = 0$ use the Ising category with positive dimension $\sqrt{2}$ and Frobenius–Schur sign $+1$. In the conventions of [1, Section 2.1, equations (9)–(13)], choose $A = e^{3\pi i/8}$; then $d = -A^2 - A^{-2} = \sqrt{2}$ and $A^4 = -i$. The displayed fermion half-braiding and pivotal normalizations give (6.6).

For $m = 2$ use unitary half-E_6, the category with (6.1). The local data in [1, Section 7.1, equation (235), Section 7.2, equation (240), Appendix D.1, equation (467)] give the normalized action square one, pivotal rotation scalar one, and the half-braiding scalar $-i$ on the noninvertible object. They therefore give exactly the second input needed in (6.6). Categorical

existence and unitarity are those of the Ising and half-E_6 constructions, not inferences from these few scalar identities alone.

The code reconstructs (6.7)–(6.10) for the three pairs, checks every fusion-ring axiom, and matches all 216 coefficients under unit-fixing permutations. For $(2,2)$ the census basis is $(1,u,v,g,a,b)$, giving the stated identification. □

The two half-E_6 factors must not be replaced without comment by a category and its complex conjugate. Conjugation changes $-i$ to $+i$; the product of the rotation phases is then $+1$, and Lemma 6.7 makes the two resolutions self-dual. The relative-product description of the 3311 category in [25, Section 5.1] uses the opposite/conjugate factors. Our non-self-dual resolution is therefore not a dimension-only reidentification of that model. The construction uses central boson condensation with a fixed point, not the earlier free-orbit formula.

6.6 The exceptional G_2 level-four model

The cyclic-four tensor $R^{(6)}_{2,4}$ has a unitary realization. We identify it with the exceptional quantum subgroup $\mathcal{E}_4(G_2)$ associated with the conformal embedding

$$G_{2,4} \ \subset \ \mathrm{Spin}(14)_1. \tag{6.11}$$

The embedding, vacuum algebra, annular matrix and branching matrix used below are given in [8, Section 4.5, pp. 29–32]. We use existence of this conformal-embedding algebra, not existence inferred merely from a commuting modular-invariant matrix.

Let $\mathcal{B}$ be the unitary WZW category of G_2 at level four, with the long fundamental weight first, as in [8]. The vacuum algebra is

$$A = V_{(0,0)} \oplus V_{(0,3)}, \qquad \dim A = 6 + 2\sqrt{6}.$$

It is the connected commutative separable algebra, with its unitary structure, supplied by (6.11). Thus $\mathcal{B}_A$, the category of all right modules with tensor product over A, is an ordinary unitary fusion category. Its subcategory of local modules is the level-one category of Spin(14). This distinction between all modules and local modules matters: the former has six simples, the latter only four.

Proposition 6.10. *The Grothendieck ring of $\mathcal{B}_A$ is based-isomorphic to $R^{(6)}_{2,4}$. In particular this ring has a unitary categorification.*

Proof. In the source ordering $(1, X, Y, h, g, g^{-1})$, the basic annular matrix and branching matrix are

$$N_X = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 2 & 2 & 1 & 0 & 0 \\ 0 & 2 & 2 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{pmatrix}, \qquad E = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{pmatrix}.$$

The source weights, indexing the rows of E, are

$$(0,0), (0,1), (0,2), (1,0), (0,3), (1,1), (0,4), (1,2), (2,0).$$

The free-module image of $V_{(0,1)}$ is the simple X, by the second row of E, so its annular action really is multiplication by X. The free-module functor is central, hence X commutes with

every object. The local simples are $1, h, g, g^{-1}$, corresponding to the vacuum, vector, and two spinor representations. The level-one fusion group of D_7 is $P/Q \simeq C_4$; equivalently its vector is the square of either spinor. Thus $g^2 = h$ and $g^4 = 1$. These four objects remain distinct and invertible in $\mathcal{B}_A$, since the local subcategory is a full tensor subcategory.

Reading the displayed matrix gives

$$X^2 = 1 + h + 2X + 2Y, \qquad hX = X, \qquad gX = g^{-1}X = Y.$$

The image X is self-dual, since $V_{(0,1)}$ is self-dual. Moreover $Y = gX$, so $Y^* = Xg^{-1} = Y$. The category is tensor generated by the central object X and the cyclic local group, so its ring is commutative. Associativity now determines all remaining products:

$$\begin{aligned} g^4 = 1, \quad gX = Y, \quad gY = X, \\ X^2 = Y^2 = 1 + g^2 + 2X + 2Y, \qquad XY = g + g^3 + 2X + 2Y. \end{aligned} \tag{6.12}$$

These are the required full fusion rules, with census order $(1, g, g^{-1}, X, Y, h)$; in source coordinates the permutation is $(0, 4, 5, 1, 2, 3)$. The resulting 216 coefficients, including the nontrivial duality $g^* = g^{-1}$ and the self-duality of X, Y, agree with $R^{(6)}_{2,4}$. □

The mathematical inputs to this identification and its arithmetic checks are separated. Existence and unitarity come from the conformal embedding. The source annular and branching matrices determine the module action; the cyclic local group then determines its full fusion ring. Independently, the constructor calculates all nine $G_{2,4}$ fusion matrices by Weyl alternants in $\mathbb{Q}(\zeta_{24})$, and verifies

$$N_i^{\mathcal{B}} E = E\Big(\sum_a E_{ia} N_a^{\mathcal{B}_A}\Big), \qquad EE^{\mathsf{T}} = I + N^{\mathcal{B}}_{(0,3)}.$$

The latter is free-module adjunction for the stated vacuum algebra. All entries, all unit-fixing identifications with the census, and the cyclic group action are exact. A dimension match alone would not prove the proposition: it would fail to distinguish the cyclic-four ring from the Klein-four rings with the same dimension list.

The dimensions are $(1, 1, 1, 1, 2 + \sqrt{6}, 2 + \sqrt{6})$ and $\mathrm{FPdim}(\mathcal{B}_A) = 24 + 8\sqrt{6}$. The published construction and its exact tensor identification provide a positive control for the finite necessary systems in Section 7.2.

6.7 Inversion equivariantization of an odd near-group category

The positive case at multiplicity three is an existing unitary model, not a proposed solution of new pentagon equations. We use the coherent order-two action constructed in [23, Section 13.3]. The following formulation separates that categorical input from the reconstruction of the entire fusion tensor.

Proposition 6.11 (An inversion-equivariantization constructor)**.** *Let n be odd, and let $\mathcal{N}$ be a unitary near-group category with invertibles α_j, $j \in \mathbb{Z}/n\mathbb{Z}$, and one further simple ρ, with*

$$\alpha_j \rho = \rho \alpha_j = \rho, \qquad \rho^2 = \bigoplus_j \alpha_j \oplus n\rho.$$

Assume a coherent unitary C_2-action fixes ρ, takes α_j to α_{-j}, fixes the unit channel of ρ^2, and acts on a chosen basis $(T_j)_j$ of $\mathrm{Hom}(\rho, \rho^2)$ by $T_j \mapsto T_{-j}$. Assume moreover that the action on the pointed part has the standard inversion equivariantization $\mathrm{Rep}(\widehat{C}_n \rtimes C_2)$. Then $\mathcal{N}^{C_2}$ is a unitary fusion category of rank $(n + 7)/2$. Its simples are

$$1, g, x_+, x_-, v_1, \ldots, v_{(n-1)/2}.$$

Writing $v_0 = 1 + g$ and $v_j = v_{-j}$, with indices modulo n, its products are determined by

$$g^2 = 1, \qquad gx_+ = x_-, \qquad gx_- = x_+, \qquad gv_i = v_i.$$

$$\begin{aligned} v_i v_j &= v_{i+j} + v_{i-j}, \\ v_i x_\pm = x_\pm v_i &= x_+ + x_-, \\ x_+^2 = x_-^2 &= 1 + \sum_i v_i + \frac{n+1}{2} x_+ + \frac{n-1}{2} x_-, \\ x_+ x_- = x_- x_+ &= g + \sum_i v_i + \frac{n-1}{2} x_+ + \frac{n+1}{2} x_-. \end{aligned}$$

All its simples are self-dual. The dimensions are $1, 1, d, d, 2, \ldots, 2$, where $d > 0$ satisfies $d^2 = n + nd$.

Proof. For an action on a semisimple category, equivariant simple objects are obtained from orbits of simple objects and irreducible projective representations of their stabilizers. In the present explicitly given action, the fixed simples $1, \rho$ have their indicated equivariant structures and each admits the two sign choices. Denote these by $1, g$ and x_+, x_-. Each nontrivial inversion orbit has trivial stabilizer and yields a single v_i with underlying object $\alpha_i \oplus \alpha_{-i}$. This proves the rank and dimension statements. Tensoring with g changes the sign of an equivariant structure and therefore exchanges x_+, x_- while fixing every v_i.

The pointed equivariantization hypothesis gives the displayed products of the v_i. For $v_i\rho$, the two summands of the underlying object are exchanged by the action. Its $+1$ and -1 eigenspaces are both one-dimensional, proving $v_i x_\pm = x_+ + x_-$, and likewise for products in the opposite order.

The unit channel in ρ^2 contributes 1 to x_+^2. The other invertible channels are paired and contribute the v_i. In the ρ-multiplicity space, inversion fixes exactly T_0 and pairs the other $n - 1$ basis vectors. Its eigenspaces have dimensions $(n + 1)/2$ and $(n - 1)/2$. This proves the formula for x_+^2. Multiplying by g proves the mixed product, and multiplying twice proves $x_-^2 = x_+^2$. The unit coefficients in these formulas show self-duality. A coherent unitary action has a unitary equivariantization; thus the construction uses actual associativity data, not merely an automorphism of a based ring. □

Corollary 6.12. *The ring $R^{(6)}_{3,218}$ admits a unitary categorification.*

Proof. Izumi's $C_5 + 5$ category has the inversion symmetry of [23, Example 13.8]. Section 13.3 constructs an actual order-two automorphism commuting with ρ, not just an outer automorphism class. Its action on the isometries is $S_j \mapsto S_{-j}$ and $T_j \mapsto T_{-j}$. Theorems 13.17–13.18 of that source establish precisely the hypotheses and product decompositions used above, including the pointed representation subcategory.

The source scalar symmetries can also be checked directly: with $\zeta = \exp(2\pi i/5)$, the quadratic phases are $a(j) = \zeta^{2j^2}$ and the nonzero b-phases have exponents $(4, 1, 1, 4)$ at $j = 1, 2, 3, 4$, each divided by $\sqrt{5}$. Both lists are inversion invariant. The dimension used here is derived from the fusion equation,

$$d = \frac{5 + 3\sqrt{5}}{2}, \qquad d^2 = 5 + 5d, \qquad \dim(\mathcal{N}^{C_2}) = 45 + 15\sqrt{5}.$$

Thus no independently copied numerical dimension enters the constructor. The source supplies existence and the coherent action; the finite checks validate the specified permutations and scalar symmetries.

In the proposition's order $(1, g, x_+, x_-, v_1, v_2)$, the census order is $(1, g, x_-, x_+, v_1, v_2)$. The function `inversion_rules(5)` constructs every product and the recorded permutation $(0, 1, 3, 2, 4, 5)$ identifies all 216 structure constants with $R^{(6)}_{3,218}$. This is a based-ring identification, not a match of Frobenius–Perron dimensions alone. □

The generic formula is not a claim that the categorical input exists for every odd n. The positive registry supplies only the cited $n = 5$ input. This constructor and its full based isomorphism are included in Table 3 and in the electronic model catalogue.

6.8 Higher subfactor and affine-group models

6.8.1 The six-object Asaeda–Haagerup category

The unitary category $\mathcal{AH}_1$ has the six-object fusion table in [22, Table 1]. In its source order $(1, \psi, \chi, \sigma, \zeta, \tau)$, the permutation $(0, 1, 3, 2, 5, 4)$ identifies all 216 structure constants with $R^{(6)}_{6,1764}$. The supplement encodes the source table separately from the census and checks this complete equality. Existence comes from the published subfactor category, not from the arithmetic of that table. This is a known unitary model. Its global dimension coinciding with that of another tensor, in particular $R^{(6)}_{3,215}$, proves neither a based-ring isomorphism nor a Morita equivalence.

6.8.2 A regular-orbit representation-ring constructor

Proposition 6.13. *Let V be a nontrivial finite abelian group and suppose that H acts freely and transitively on $\widehat{V} \setminus \{1\}$. Put $h = |H| = |V| - 1$. The irreducibles of $G = V \rtimes H$ consist of the inflations of* $\mathrm{Irr}(H)$ *and one additional self-dual representation ρ of dimension h. Their products satisfy*

$$\sigma\rho = (\dim \sigma)\rho, \qquad \rho^2 = \sum_{\sigma \in \mathrm{Irr}(H)} (\dim \sigma)\sigma + (h-1)\rho. \tag{6.13}$$

Consequently these rules, together with $\mathrm{Rep}(H)$*, define a unitary categorical model, whenever the stated group action is supplied.*

Proof. The irreducibles of a semidirect product by an abelian normal subgroup are obtained from H-orbits in $\widehat{V}$ and representations of their stabilizers. The trivial orbit gives the inflated representations of H. The single nontrivial orbit has trivial stabilizer and gives $\rho = \mathrm{Ind}_V^G \lambda$, for any nontrivial λ. Inversion preserves this orbit, so ρ is self-dual. Restriction of an inflated σ to V is $(\dim \sigma)1$, and induction gives the first equation. Its reciprocity then gives the multiplicity $\dim \sigma$ of each inflation in ρ^2. If the remaining multiplicity of ρ is a, comparing dimensions yields $h^2 = \sum_\sigma (\dim \sigma)^2 + ah = h + ah$, so $a = h - 1$. The representation category of a finite group is unitary. □

Take $V = \mathbb{F}_3^2$ and let H be generated by

$$I = \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix}, \qquad J = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

The identities $I^2 = J^2 = -1$ and $IJ = -JI$ identify H with Q_8. Its orbit on the eight nonzero vectors is free and transitive; the contragredient action has the same property. Thus Proposition 6.13 gives a six-object category of dimension 72, with simple dimensions $(1, 1, 1, 1, 2, 8)$ and multiplicity seven. It realizes $R^{(6)}_{7,572}$.

The constructor generates the actual 72-element semidirect product, checks its multiplication and action, and computes the four linear characters, the inflated two-dimensional

quaternion character, and the induced eight-dimensional character. It verifies orthogonality, completeness by the sum of squares, and all tensor coefficients by character inner products. The resulting tensor is then compared to the census. This test is an application of the proposition to explicit group data, not a rule accepting all abstract rings with these dimensions.

6.9 The complete catalogue and group-theoretical constructions

The positive part of the classification is recorded as explicitly as the negative part. Table 3 contains one row for every one of the 54 positive rings, including all twenty-one rings of multiplicity one. The electronic catalogue records every constructor used, not merely one representative: there are 61 constructor records, with an explicit unit-fixing basis permutation from each output to its census representative. These permutations are verified entry by entry. The names in the table are presentation labels, not hypotheses in a categorification test. The symbols U and S distinguish unitary existence from spherical existence with proved nonexistence of a pseudo-unitary categorification. The single S entry is not counted among the unitary models.

6.9.1 Small constructions and conventions

Here $PSU(2)_k$ means the even subcategory of the principal unitary $SU(2)_k$ category, and D_9 has order eighteen. The indicated C_2 action on $C_3 \times C_3$ is inversion. In $C_3 \rtimes C_4$, a generator of C_4 acts by inversion on C_3; the action therefore factors through C_2. The positive Ising-type rings in the table are denoted $\mathcal{I}(G, r)$, where $|G| = 4$, a subgroup H of order two is fixed, and $r \in \{0, 1\}$. Their basis consists of G and σ_0, σ_1. An element of $G/H \cong C_2$ translates the subscripts, and

$$\sigma_i \sigma_j = \sum_{a \in G:\, \bar{a} = i+j+r} a.$$

The unitary realizations are the Ising zestings constructed in [27, Section 4.3]; the metaplectic category $SO(5)_2$ and its listed zest are treated there as well. The program reconstructs these formulas, ordinary finite-group character products, the Haagerup tensor, and the $SU(2)$ fusion rules before matching the resulting tensors. Thus the multiplicity-one positives do not depend on a cardinality-complement argument.

Table 3: All positively identified rings in the census. U means a unitary realization; S means a spherical realization and no pseudo-unitary categorification (hence no unitary one). The numbers of types U and S are given in Table 1. Here $R_{m,j} = R^{(6)}_{m,j}$. Every constructor and its based isomorphism are recorded electronically.

Ring	Model or construction	Type	Source
$R_{1,1}$	$\mathcal{I}(C_4, 1)$: cyclic, non-self-dual Ising zest	U	[27]
$R_{1,3}$	$\mathcal{I}(C_4, 0)$: cyclic, self-dual Ising zest	U	[27]
$R_{1,4}$	$\mathrm{Rep}(C_3 \rtimes C_4)$	U	[27]
$R_{1,6}$	$\mathrm{Vec}_{C_2} \boxtimes \mathrm{Vec}_{C_3} \simeq \mathrm{Vec}_{C_6}$	U	[27]
$R_{1,7}$	Tambara–Yamagami category for C_5	U	[27]
$R_{1,9}$	$\mathcal{I}(C_2^2, 1)$: non-self-dual Ising zest	U	[27]
$R_{1,10}$	Zested $SO(5)_2$ category	U	[27]
$R_{1,11}$	$\mathrm{Fib} \boxtimes \mathrm{Vec}_{C_3}$	U	[27]
$R_{1,14}$	Vec_{S_3}	U	[27]
$R_{1,15}$	Haagerup category $\mathcal{H}_3$	U	[21]
$R_{1,20}$	$\mathrm{Vec}_{C_2} \boxtimes$Ising; also $R(0, 0)$	U	[27, 28]

Continued on the next page

Table 3 continued

Ring	Model or construction	Type	Source
$R_{1,21}$	$\mathrm{Vec}_{C_2} \boxtimes \mathrm{Rep}(S_3)$; also $R(0,1)$	U	[27, 28]
$R_{1,23}$	$\mathrm{Vec}_{C_2} \boxtimes PSU(2)_5$	U	[27]
$R_{1,24}$	$PSU(2)_{10}$ (the other 3311 even-part ring)	U	[27, 25]
$R_{1,26}$	$SO(5)_2$	U	[27]
$R_{1,30}$	$\mathrm{Rep}(D_9)$, with $\lvert D_9\rvert = 18$	U	[27]
$R_{1,32}$	$\mathrm{Rep}((C_3 \times C_3) \rtimes C_2)$, inversion action	U	[27]
$R_{1,34}$	$PSU(2)_{11}$	U	[27]
$R_{1,36}$	$\mathrm{Fib} \boxtimes \mathrm{Ising}$	U	[27]
$R_{1,37}$	$\mathrm{Fib} \boxtimes \mathrm{Rep}(S_3)$	U	[27]
$R_{1,38}$	$\mathrm{Fib} \boxtimes PSU(2)_5$	U	[27]
$R_{2,2}$	Fermionic C_2-zest of $\mathrm{Ad}(E_6) \boxtimes \mathrm{Vec}_{C_2}$	U	[9]
$R_{2,4}$	$\mathcal{E}_4(G_2)$: modules for $G_{2,4} \subset \mathrm{Spin}(14)_1$	U	[8]
$R_{2,11}$	$\mathrm{Rep}((C_3 \times C_3) \rtimes C_4)$	U	[15]
$R_{2,14}$	Fermionic grading zest of the E_6 category	U	[9]
$R_{2,34}$	TT_3	U	[13]
$R_{2,36}$	$\mathcal{C}(A_4, 1, C_2, 1)$	U	[2]
$R_{2,40}$	Diagonal-fermion quotient of two identical half-E_6 categories	U	[1]
$R_{2,49}$	Quadratic category $R'(2)$	U	[28]
$R_{2,53}$	$\mathrm{Vec}_{C_2} \boxtimes \mathrm{Ad}(E_6)$; also $R(0,2)$	U	[28]
$R_{2,69}$	C_2-extension of $\mathrm{Rep}(H_8)$	U	[7]
$R_{2,71}$	$\mathcal{C}(S_4, \omega, C_2^2, \psi)$ of Theorem 6.16	U	[20]
$R_{2,75}$	E_6 modules over $SU(2)_{10}$	U	[26]
$R_{2,94}$	Diagonal-fermion quotient of $\mathrm{Ising} \boxtimes PSU(2)_6$	U	[15]
$R_{2,97}$	Dual Goodman–de la Harpe–Jones 3311 even part	U	[25]
$R_{2,98}$	Dual even part of the Izumi C_4 subfactor	U	[24]
$R_{2,105}$	$\mathcal{C}(A_5, 1, D_{10}, 1)$	U	[2]
$R_{2,124}$	Schopieray: $\mathcal{C}(\mathfrak{so}_5, 3/2)_{\mathrm{ad}}$	S	[41]
$R_{2,127}$	$\mathrm{Fib} \boxtimes \mathrm{Ad}(E_6)$	U	[26]
$R_{2,128}$	D_6 modules over $SU(2)_8$	U	[26]
$R_{2,140}$	Local D_{10} modules over $SU(2)_{16}$	U	[26]
$R_{2,143}$	Commutative even part of the Izumi C_5 subfactor	U	[17]
$R_{2,144}$	Principal G_2 level-three category	U	[40]
$R_{3,47}$	$\mathrm{Rep}(GL(3, \mathbb{F}_2)) \simeq \mathrm{Rep}(PSL(2,7))$	U	[2]
$R_{3,141}$	$\mathcal{C}(A_5, 1, S_3, 1)$	U	[2]
$R_{3,218}$	Inversion equivariantization of Izumi's $C_5 + 5$ near-group category	U	[23]
$R_{4,249}$	Sign-swap equivariantization of the equal half-E_6 quotient	U	§6.12
$R_{4,322}$	Diagonal-fermion quotient of half-$E_6 \boxtimes PSU(2)_6$	U	§6.11
$R_{5,17}$	Izumi's $C_5 + 5$ near-group category	U	[23]
$R_{6,345}$	$\mathcal{C}_{4,8,4}$, with ring $R(4,6)$	U	[28]
$R_{6,1712}$	Relative Fibonacci center $Z_{\mathrm{Fib}}(TT_3)$	U	§6.13
$R_{6,1764}$	Asaeda–Haagerup category $\mathcal{AH}_1$	U	[22]
$R_{7,572}$	$\mathrm{Rep}((C_3 \times C_3) \rtimes Q_8)$	U	[15]

Continued on the next page

Table 3 continued

Ring	Model or construction	Type	Source
$R_{7,2274}$	Extended Haagerup category $\mathcal{EH}_1$	U	[4, 6]

6.9.2 Exact Weyl-character reconstruction

We use the principal unitary categories $\mathcal{C}(\mathfrak{g}, k)$ at positive integral level. Their construction and the Weyl-alcove and character formulas are reviewed in [40]; these are categorical inputs, not consequences of a numerically fitted modular matrix. Normalize long roots to have squared length two, let W be the Weyl group, and put $\rho = \sum_i \lambda_i$. For dominant weights in the level-k alcove define

$$C_{\lambda\mu} = \frac{\sum_{w\in W} \det(w) \exp\bigl(-2\pi i (w(\lambda+\rho), \mu+\rho)/(k+h^\vee)\bigr)}{\sum_{w\in W} \det(w) \exp\bigl(-2\pi i (w\rho, \mu+\rho)/(k+h^\vee)\bigr)}. \tag{6.14}$$

These are the fusion characters. Their products determine the fusion coefficients by

$$C_{\lambda\mu} C_{\nu\mu} = \sum_\tau N_{\lambda\nu}^\tau C_{\tau\mu}.$$

All denominators and the required character-matrix rank are checked in the cyclotomic number field. Exact Gaussian elimination determines the coefficients, which are then checked to be nonnegative integers and to satisfy every character equation, including columns not used to solve the linear system. This also works for the root-lattice subcategory of type A_2, using a full-rank submatrix of its restricted characters.

For G_2 at level three, the short fundamental weight is first, and the input data are

$$((\lambda_i, \lambda_j)) = \begin{pmatrix} 2/3 & 1 \\ 1 & 2 \end{pmatrix}, \qquad h^\vee = 4, \qquad \Lambda_3 = \{(0,0), (1,0), (2,0), (3,0), (0,1), (1,1)\}.$$

The calculation takes place in $\mathbb{Q}(\zeta_{21})$, and produces a rank-six tensor of multiplicity two.

Proposition 6.14. *The tensor obtained from* (6.14) *for* $(\mathfrak{g}, k) = (G_2, 3)$ *is based-isomorphic to* $R^{(6)}_{2,144}$. *In particular this ring admits a unitary modular categorification.*

Proof. The Weyl construction supplies the category and the displayed character formula supplies its fusion rules. The exact calculation just described is implemented independently of the census in `software/quantum_rank2.py`. The resulting 216 coefficients and a basis permutation are recorded in the model catalogue; the checker recomputes them and verifies their equality to the census tensor. This is a full-tensor identification, not an identification by dimension type. □

6.9.3 The dual 3311 even part

The dual even-part identification is

$$R^{(6)}_{2,97} \cong K_0\bigl((A_{11}^{\mathrm{even}})^*_{E_6^{\mathrm{even}}}\bigr). \tag{6.15}$$

The category on the right is the dual even part of the Goodman–de la Harpe–Jones 3311 subfactor and is unitary. Its complete multiplication table is Figure 2 of [25, Section 5.1]. The independent constructor uses the basis $(1, B, C, C', D, g)$ of that figure; in particular

$$C^2 = 1 + g + 2C, \quad (C')^2 = 1 + g + 2C', \quad CC' = B + D, \quad gC = C, \quad gC' = C', \quad gB = D.$$

The other products are reconstructed from the same published table and all tensor axioms and the based isomorphism in (6.15) are checked. The other 3311 even-part ring is already $R^{(6)}_{1,24}$. Neither identification implies that another ring with the same Frobenius–Perron dimensions is categorifiable; in particular it does not settle the non-self-dual ring $R^{(6)}_{2,40}$.

6.9.4 A three-double-coset realization criterion

For finite G, a subgroup H, and unit-modulus normalized cochains $\omega \in Z^3(G,\mathbb{C}^\times)$ and $\psi \in C^2(H,\mathbb{C}^\times)$ with $\omega|_H = \delta\psi$, write $\mathcal{C}(G,\omega,H,\psi)$ for the category of ψ-twisted H-bimodules in Vec_G^ω, with the conventions of [20]. For $u \in G$ put $H^u = H \cap uHu^{-1}$. Simple objects over HuH correspond to irreducible projective representations of H^u with factor

$$\psi^u(a,b) = \psi(a,b)\psi(u^{-1}b^{-1}u, u^{-1}a^{-1}u) \cdot \frac{\omega(a,b,u)\omega(a,bu,u^{-1}b^{-1}u)}{\omega(abu,u^{-1}b^{-1}u,u^{-1}a^{-1}u)}. \tag{6.16}$$

Such a simple has Frobenius–Perron dimension $[H:H^u]\dim\pi$. These statements, and support of tensor products on products of double cosets, are [20, Theorems 4.1 and 5.1]. We call a factor on an abelian group nondegenerate when its twisted group algebra has one irreducible representation; equivalently its commutator bicharacter has trivial radical.

Theorem 6.15 (Three-double-coset realization)**.** *Suppose H is abelian of order $h = s^2 > 1$ and*

$$G = H \sqcup tH \sqcup HuH, \qquad N_G(H) = H \sqcup tH, \qquad H^u = \{1\}.$$

If the factor ψ^t in (6.16) *is nondegenerate, then $\mathcal{C}(G,\omega,H,\psi)$ is a unitary categorification of the commutative fusion ring with basis $\widehat{H} \sqcup \{X,Y\}$ and rules*

$$\chi X = X\chi = X, \qquad \chi Y = Y\chi = Y, \qquad XY = YX = sY,$$
$$X^2 = \sum_{\chi\in\widehat{H}} \chi, \qquad Y^2 = \sum_{\chi\in\widehat{H}} \chi + sX + (h-2)Y. \tag{6.17}$$

Its dimension type is $(1^h, s, h)$.

Proof. The component over H is $\mathrm{Rep}(H)$, so it consists of the h invertibles indexed by $\widehat{H}$ with their ordinary group multiplication. Nondegeneracy of ψ^t gives one simple X of dimension s over tH. The trivial stabilizer of HuH gives one simple Y of dimension h. Hence these are all the simple objects.

The normalizer is a subgroup with quotient $N_G(H)/H$ of order two. Consequently tH is closed under inversion. Its complement HuH in G is also closed under inversion and is stable under multiplication on either side by $N_G(H)$. Since X and Y are unique on their supports, both are self-dual and all the invertibles fix them on both sides. Support forces X^2 into $\mathrm{Rep}(H)$. Frobenius reciprocity gives $[X^2:\chi] = [\chi X : X] = 1$, proving the first square formula. Similarly, XY and YX are supported on HuH, and dimensions give $XY = YX = sY$. Reciprocity gives $[Y^2:\chi] = 1$ and $[Y^2:X] = s$. The remaining coefficient is therefore determined by $h^2 = h + s^2 + h[Y^2:Y]$, namely $[Y^2:Y] = h-2$.

Finally Vec_G^ω is unitary for unit-modulus ω. The admissible twisted regular algebra of H, with its normalized regular inner product, is a connected separable C^*-algebra object. Its bimodule category is unitary. This is the standard unitary form of the group-theoretical construction, and proves existence rather than merely associativity of the displayed ring. □

6.9.5 An explicit twisted S_4 model

The following cochain certificate makes the realization criterion applicable to a residual that is not realized by the corresponding untwisted construction.

Realize $G = PGL(2,\mathbb{F}_3) \cong S_4$ by its action on $\mathbb{P}^1(\mathbb{F}_3) = \{(1,0),(0,1),(1,1),(1,2)\}$. In each projective class choose the matrix $s(g)$ whose first nonzero entry in row-major order is

one. Define

$$s(g)s(k) = (-I)^{\beta(g,k)}s(gk), \qquad \alpha(g) = \begin{cases} 0, & \det s(g) = 1, \\ 1, & \det s(g) = 2, \end{cases} \qquad \omega(g,k,l) = (-1)^{\alpha(g)\beta(k,l)}.$$

Here β is a normalized $\mathbb{F}_2$-valued 2-cocycle from the central extension $GL(2,\mathbb{F}_3)$, and α is a homomorphism. The cup-product identity therefore proves that ω is a normalized 3-cocycle. Take $H = \{1, a, b, c\}$ represented, respectively, by

$$I, \quad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}.$$

Its action is the subgroup generated by the two disjoint transpositions on the displayed ordered projective line, so $H \cong C_2^2$. Put $\psi(x,y) = i^{q(x,y)}$, with $q(1,x) = q(x,1) = 0$ and

q	a	b	c
a	0	2	0
b	3	0	1
c	1	0	1

in $\mathbb{Z}/4\mathbb{Z}$. (6.18)

For every $x, y, z \in H$ the admissibility identity is

$$q(y,z) + q(x,yz) - q(xy,z) - q(x,y) \equiv 2\alpha(x)\beta(y,z) \pmod 4.$$

It follows either by substitution in the nine-entry table or by the complete 4^3-triple check in the supplement.

Theorem 6.16. *For the cochains just defined, $\mathcal{C}(S_4, \omega, C_2^2, \psi)$ is a unitary categorification of $R_{2,71}^{(6)}$.*

Proof. Choose

$$t = \left[\begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}\right], \qquad u = \left[\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}\right].$$

Matrix multiplication over $\mathbb{F}_3$ gives exactly three double cosets, of sizes $4, 4, 16$, represented by $1, t, u$. The first two form $N_G(H)$, and H^u is trivial. Substitution in (6.16) gives the following exponent matrix for ψ^t, in the order $(1, a, b, c)$:

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 3 & 0 & 1 \\ 0 & 2 & 3 & 1 \end{pmatrix} \pmod 4.$$

Its commutator bicharacter has trivial radical: each nonidentity row has an entry different from the corresponding transposed entry. Thus its twisted group algebra is $M_2(\mathbb{C})$, and Theorem 6.15 applies with $h = 4$, $s = 2$.

The constructor regenerates the twenty-four projective matrices, their faithful permutation action, the subgroup, all double cosets, and every stabilizer factor. It checks the admissibility identity and, in addition to the cup-product proof, all 24^4 cocycle identities. It then uses (6.17) to reconstruct the full tensor and checks its based isomorphism with $R_{2,71}^{(6)}$. The complete data are in `data/twisted_s4_certificate.json`. Thus no inference from Frobenius–Perron dimensions alone is used. □

The three additions in this section are positive determinations, not new exclusions. In particular, searching only untwisted group-theoretical models would miss the last construction. All allocated negative certificates remain applications of the unified obstruction framework stated earlier in the paper.

6.10 Exceptional and near-group source identifications

Two source tensors must be included independently of any obstruction search. The principal even part $\mathcal{EH}_1$ of the extended Haagerup subfactor is unitary by its subfactor construction [4]. Its six fusion matrices are explicitly displayed in [6, Section 5.1]. In that source order, the permutation $(0,1,3,4,5,2)$ gives $R^{(6)}_{7,2274}$. The largest coefficient is seven. The constructor transcribes the published matrices and checks all 216 coefficients, not merely the dimension field or formal codegrees.

The unitary C_5+5 near-group category in [23, Example 13.8] has basis $\alpha_0,\dots,\alpha_4,\rho$ and products

$$\alpha_i\alpha_j=\alpha_{i+j},\qquad \alpha_i\rho=\rho\alpha_i=\rho,\qquad \rho^2=\sum_{i=0}^{4}\alpha_i+5\rho.$$

Its tensor identifies $R^{(6)}_{5,17}$ with permutation $(0,1,4,3,2,5)$. The positive root of the dimension equation is $(5+3\sqrt{5})/2$. This is the near-group input itself, distinct from the inversion equivariantization of Corollary 6.12. Both source identifications are independent of the census and both are matched under all unit-preserving permutations.

6.11 A mixed diagonal-fermion quotient

Let $\mathcal{E}$ be the unitary half-E_6 category of (6.1), and let $\mathcal{P}=PSU(2)_6$. The object h of $\mathcal{E}$ has the central fermion of Lemma 6.1. In the standard unitary $SU(2)_6$ category, V_6 is an order-two fermion; restricting its half-braiding gives a central fermion in $\mathcal{P}$. Thus (h,V_6) is a central boson in $\mathcal{E}\boxtimes\mathcal{P}$.

Proposition 6.17. *The category of modules over $1\oplus(h,V_6)$ in $\mathcal{E}\boxtimes\mathcal{P}$ is a unitary fusion category with Grothendieck ring $R^{(6)}_{4,322}$.*

Proof. The second factor has basis (V_0,V_2,V_4,V_6), with V_6 interchanging V_0,V_6 and V_2,V_4. The diagonal action is therefore free, even on objects whose first component is X. Proposition 6.6 applies without a fixed-point choice. In the product order $(1,h,X)\times(V_0,V_2,V_4,V_6)$, its permutation of the twelve simples is

$$(7,6,5,4,3,2,1,0,11,10,9,8).$$

Take orbit representatives $(0,1,2,3,8,9)$. The quotient formula of Proposition 6.6, applied to the independently constructed half-E_6 and $SU(2)_6$ tensors, gives the census tensor after permutation $(0,3,1,2,4,5)$. The implementation checks every product and its independence of the chosen representative. Existence and unitarity come from the central separable algebra; the tensor comparison identifies its Grothendieck ring. □

6.12 A sign-twisted swap on a fermionic product

The next construction has a categorical hypothesis. Associativity of the integer tensor for a formal parameter does not provide its input category.

Theorem 6.18 (Sign-swap equivariantization)**.** *Let $\mathcal{E}$ be a unitary fusion category with simples $1,f,x$, a specified central fermion f, and rules*

$$f^2=1,\quad fx=xf=x,\quad x^2=1+f+mx.$$

Suppose $d=\mathrm{FPdim}(x)$ is irrational. There is a unitary fusion category with invertible group $G=C_2\times C_2$ and two self-dual simples X,Y, fixed by all invertibles, with

$$X^2=\sum_{g\in G}g+mX+2Y, \tag{6.19}$$

$$XY = YX = 2X + 2mY,$$
$$Y^2 = \sum_{g\in G} g + 2mX + m^2 Y.$$

Their dimensions are $2d$ and d^2. The Ising input $m = 0$ gives $R^{(6)}_{2,69}$, and the half-E_6 input $m = 2$ gives $R^{(6)}_{4,249}$.

Proof. In $\mathcal{E}\boxtimes\mathcal{E}$ put $b = f\boxtimes f$ and $A = 1\oplus b$. This is the regular algebra of a central Tannakian subcategory. The factor-swap tensor functor S preserves the central lift of b. Equip A with the algebra isomorphism $\phi : S(A) \to A$ that is $+1$ on 1 and -1 on b. It satisfies $\phi S(\phi) = \mathrm{id}$. Transport of module actions through S and ϕ defines a coherent unitary C_2 action T on $\mathcal{Q} = (\mathcal{E}\boxtimes\mathcal{E})_A$. The category we use is the ordinary equivariantization $\mathcal{Q}^{C_2}$.

The free b-orbits give four simples of $\mathcal{Q}$: the unit, $g = (f\boxtimes 1)A$, and $a = (x\boxtimes 1)A$, $a' = (1\boxtimes x)A$. The fixed object $x\boxtimes x$ gives two simples u_+, u_-, each of dimension $d^2/2$. To see the action explicitly, choose the right f-action $t : x\otimes f \to x$ with square one, using the module associativity constraint. Such a normalization exists because $1\oplus f$ is the regular separable algebra and its two actions on x differ by sign. The right b-actions on the fixed object are $\pm(t\boxtimes t)$. Swap preserves this tensor, while ϕ changes its sign. Hence T interchanges u_+, u_-; it also interchanges a, a' and fixes $1, g$.

An orbit of size two has trivial stabilizer and yields one simple in the equivariantization; a fixed simple has two equivariant structures, since projective representations of C_2 over $\mathbb{C}$ can be linearized. Thus there are four invertibles, one simple X over $a\oplus a'$ and one simple Y over $u_+\oplus u_-$. Their dimensions are $2d, d^2$. Their supports are invariant under duality, so X, Y are self-dual. They are fixed by all invertibles: tensoring preserves their distinct dimensions and there is only one simple of either dimension.

We check the group law on the four invertibles; dimensions alone would not distinguish C_4 from $C_2\times C_2$. Work in the pointed part $\mathrm{sVec}\boxtimes\mathrm{sVec}$, with trivial associator and the specified self-braidings -1. Let e_0, e_1 be the two homogeneous components of $g = (f, 1)A$, in degrees $(f, 1), (1, f)$. An isomorphism $g\otimes_A g \to A$ has the normalized coefficients

$$\mu_{00} = 1, \qquad \mu_{01} = b, \qquad \mu_{10} = -b, \qquad \mu_{11} = -1.$$

The minus sign in balancing is forced by the half-braiding of b with $(f, 1)$; it is not a choice of a ring sign. Give g the equivariance $T(e_0)\mapsto ie_1$, $T(e_1)\mapsto -ie_0$. Its square is the identity. On each of the four pairs (e_r, e_s), the displayed μ intertwines this tensor-square equivariance with ϕ, which acts on A as $(1, -1)$. Therefore this lift of g squares to the tensor unit. Together with the independent sign representation of C_2, it generates $C_2\times C_2$. The six scalar equalities (two for coherence, four for multiplication) are also checked exactly over $\mathbb{Q}(i)$.

Frobenius reciprocity now puts each invertible once in X^2, Y^2 and none in XY. Write $d^2 = 2+md$. Comparing coefficients of $1, d$ in $(2d)^2$, $(2d)d^2$ and d^4 forces respectively the coefficients in (6.19); their uniqueness uses the irrationality of d. For example $X^2 = \sum_g g + aX + cY$ gives $8 + 4md = 4 + 2c + (2a + cm)d$, so $c = 2$ and $a = m$. For $XY = aX + cY$, comparison gives $c = 2m, a = 2$; for $Y^2 = \sum_g g + aX + cY$, it gives $c = m^2, a = 2m$. All operations used preserve unitarity.

The Ising and half-E_6 inputs and their central fermions are supplied by standard Ising theory and Lemma 6.1 (see also [1]). Their tensors, constructed from (6.19), match the census with permutations $(0, 1, 2, 3, 5, 4)$ and the identity. These two categorical inputs, not arbitrary values of m, are admitted by the model registry. □

6.13 Reconstruction from a relative center

For a full fusion subcategory $\mathcal{D}\subset\mathcal{C}$, the relative center $Z_{\mathcal{D}}(\mathcal{C})$ consists of objects of $\mathcal{C}$ equipped with a coherent half-braiding against objects of $\mathcal{D}$. It is not the Müger centralizer in a

braided category. It is a fusion category, is Morita equivalent to $\mathcal{D} \boxtimes \mathcal{C}^{\mathrm{rev}}$, and has dimension $\mathrm{FPdim}(\mathcal{D})\,\mathrm{FPdim}(\mathcal{C})$ [19, Remark 2.4]. For unitary input, its realization as the dual of the natural unitary module category gives a unitary fusion category.

Proposition 6.19 (Forced relative-center reconstruction)**.** *Suppose $\mathcal{D} = \langle a \rangle \subset \mathcal{C}$ is a Fibonacci subcategory, so $a^2 = 1 + a$. In the simple basis of $\mathcal{C}$, let L_a, R_a be the matrices of left and right multiplication by a, and put*

$$B = I + L_a R_a, \qquad L = (e_0, e_a, e_a, e_0 + e_a), \qquad Q = B - LL^{\mathsf{T}}.$$

Starting from Q, whenever a diagonal entry is one, its column is a forced nonnegative integral Gram column. Subtract its outer product and repeat. If this process reaches zero, it determines the entire restriction matrix $F = (L, H)$ of $Z_{\mathcal{D}}(\mathcal{C})$.

Assume additionally that the high columns are pairwise distinct, distinct from the low columns, and individually invariant under duality in $\mathcal{C}$. Then the relative center is self-dual. Impose the known $\mathrm{Fib} \boxtimes \mathrm{Fib}^{\mathrm{rev}}$ multiplication on its first four simples and the ring-homomorphism equations for F. If these linear equations in the symmetric nonunit structure constants have a unique solution, that solution is its complete Grothendieck tensor.

Proof. For the forgetful functor and its adjoint, relative central induction has underlying object $FI(V) = V \oplus aVa$; equivalently it is the finite semisimple coend $\bigoplus_{U \in \mathrm{Irr}\,\mathcal{D}} U \otimes V \otimes U^*$. Adjunction gives the Gram identity $FF^{\mathsf{T}} = B$. The full inclusion $\mathcal{D} \subset \mathcal{C}$ induces a fully faithful tensor inclusion $Z(\mathcal{D}) \subset Z_{\mathcal{D}}(\mathcal{C})$: the half-braiding condition and all morphism spaces between objects of $\mathcal{D}$ are unchanged. The standard nondegenerate Fibonacci braiding identifies $Z(\mathcal{D})$ with $\mathrm{Fib} \boxtimes \mathrm{Fib}^{\mathrm{rev}}$, whose four restriction columns are exactly L.

If $Q_{ii} = 1$, exactly one remaining simple has restriction coefficient one in row i, and all the others have coefficient zero. Its full restriction column is therefore $Q_{\bullet i}$. Subtracting its outer product is mandatory. At zero residual no further nonzero restriction column is possible. This proves the asserted uniqueness without a numerical factorization or a conjectural central rank.

The low four simples are self-dual. The additional hypotheses force every high simple to be self-dual, because duality permutes restriction columns and no other simple has the same column. A based ring all of whose simples are self-dual is commutative. Frobenius reciprocity makes its nonunit coefficients symmetric in all three indices. Its forgetful functor is a ring homomorphism and the low multiplication is already known. Hence its actual tensor solves the stated linear system. Uniqueness determines that tensor; categorical existence was supplied before solving the system. If a pivot or uniqueness test fails, the recipe is inconclusive, not an exclusion or a claimed construction from arbitrary ring data. □

Corollary 6.20. *For the unitary category TT_3 and either of its Fibonacci generators, $Z_{\mathrm{Fib}}(TT_3)$ has Grothendieck ring $R^{(6)}_{6,1712}$.*

Proof. Use the source basis $(1, f^{(2)}, \rho, \sigma, \bar{\sigma}, \mu)$ of [13, Definition 2.5] and $a = \rho$. Its dimension vector is $(1, \varphi^3, \varphi, \varphi^2, \varphi^2, \varphi)$, where $\varphi = (1 + \sqrt{5})/2$. The source provides the unitary category and its complete multiplication table. The Gram procedure of Proposition 6.19 gives

$$F = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

Indeed, after removal of the low columns the row of σ is a diagonal-one pivot; after its forced subtraction the row of $f^{(2)}$ is another such pivot. Their removal leaves zero. Denote the

columns by $(1, p, q, pq, X, Y)$. Then $p^2 = 1 + p$, $q^2 = 1 + q$, and the complete remaining rules are

$$\begin{aligned} pX = qX = Y, \qquad & pY = qY = X + Y, \\ (pq)X = X + Y, \qquad & (pq)Y = X + 2Y, \\ X^2 = 1 + pq + 2X + 2Y, & \\ XY = p + q + pq + 2X + 4Y, & \\ Y^2 = 1 + p + q + 2pq + 4X + 6Y. & \end{aligned}$$

For detail on uniqueness, there are $\binom{7}{3} = 35$ symmetric nonunit unknowns. The low products and all 36 restriction-product identities give 312 scalar linear equations of exact rank 35. The verifier reconstructs this integer system and its unique nonnegative integral solution, and independently checks every based-ring axiom. The displayed rules also permit direct substitution into every identity.

The full tensor matches $R^{(6)}_{6,1712}$ under $(0, 1, 2, 4, 5, 3)$. Repeating the calculation with $a = \mu$ gives a based-isomorphic result. Applying the same recipe to $\mathcal{C} = \mathcal{D} = \mathrm{Fib}$ recovers $\mathrm{Fib} \boxtimes \mathrm{Fib}^{\mathrm{rev}}$ as an independent positive control. Thus neither the rank-six answer nor its census identification is an input to the reconstruction. □

7 Certificates, research coverage, and the remaining frontier

7.1 Automated allocation and verification

The computation has three separate layers: the frozen source census, a discovery pipeline, and an exact categorification-certificate checker. Discovery is permitted to time out or to use numerical linear programming to propose a functional. Only the checker can validate a negative record. The final ledger contains one of four disjoint reporting classes: generally excluded, unitarily realized, spherically realized with no pseudo-unitary categorification, or unresolved. Five separate existence columns record the categorical structures, with no implication from spherical to pseudo-unitary existence. The pipeline rejects any intersection of its positive and negative sets.

Table 4 gives a sequential view: a tensor removed at one stage is not counted as a new removal at any later stage. The order places inexpensive structural and arithmetic obstructions before larger finite branch searches. Effectiveness means the number of *additional* tensors decided at that point, rather than the number that a criterion could reject when run in isolation. The table is a transparent operational order, not a comparison of logical strength or a hardware-independent timing ranking. Ineligible hypotheses, an unfinished search and a feasible necessary relaxation all leave a ring undecided unless another accepted certificate or model resolves it. At multiplicity eight, the dimension-order and character-induction stages first contribute new removals in this top-level sieve, excluding one and four additional rings respectively. This first contribution does not assert that these mathematical tools were absent from nested certificates at lower multiplicities, nor that the repertoire will suffice at multiplicity nine. Nineteen rings remain unresolved at exact multiplicity eight.

The allocation in Table 5 is disjoint. It records the certificate family stored for each exclusion, while Table 4 makes the sequential accounting explicit. Neither table asserts mathematical independence of the criteria. The signed and positive-dimension records call the same low-induction and finite-indicator framework internally. When an older unbalanced relative-trace certificate suffices, it is interpreted as a certificate for Theorem 4.3 with zero coefficients on the added balance equations. Likewise every inherited Farkas vector remains valid after the Galois-forbidden columns of Theorem 4.7 are removed: the inequalities on

Criterion	$m = 1$	$m = 2$	$m = 3$	$m = 4$	Total removed	Seconds
Input	39	154	384	872	—	—
Rank-two subring	0/39	4/150	4/380	49/823	631	0.9
d-number	10/29	87/63	312/68	687/136	14753	11.6
Zero spectrum	2/27	0/63	1/67	0/136	4	0.6
Lower-rank subring	0/27	0/63	6/61	16/120	144	0.4
Involution module	4/23	3/60	19/42	13/107	233	52.0
Dimension quotient	0/23	1/59	2/40	2/105	20	6.5
Cyclotomicity	0/23	0/59	2/38	1/104	14	2.2
Morita descent	0/23	2/57	1/37	1/103	4	0.6
Field automorphisms	0/23	1/56	0/37	0/103	2	5.8
Weighted induction	0/23	1/55	1/36	1/102	7	5.5
Sphericalization/induction	2/21	33/22	32/4	85/17	693	857.6
Dimension order	0/21	0/22	0/4	0/17	1	184.1
Character induction	0/21	0/22	0/4	0/17	4	98.7
Tensor rotation	0/21	0/22	0/4	0/17	0	81.8
Modular induction	0/21	0/22	0/4	0/17	0	65.2
Deferred exact checks	0/21	0/22	0/4	0/17	1	157.2
Constructed models	21	22	3	2	—	—
Unresolved	0	0	1	15	—	—

Criterion	$m = 5$	$m = 6$	$m = 7$	$m = 8$
Input	1582	2768	3814	6999
Rank-two subring	81/1501	121/2647	113/3701	259/6740
d-number	1356/145	2442/205	3473/228	6386/354
Zero spectrum	0/145	1/204	0/228	0/354
Lower-rank subring	25/120	28/176	22/206	47/307
Involution module	49/71	23/153	78/128	44/263
Dimension quotient	6/65	2/151	4/124	3/260
Cyclotomicity	1/64	3/148	0/124	7/253
Morita descent	0/64	0/148	0/124	0/253
Field automorphisms	0/64	1/147	0/124	0/253
Weighted induction	1/63	1/146	1/123	1/252
Sphericalization/induction	58/5	140/6	115/8	228/24
Dimension order	0/5	0/6	0/8	1/23
Character induction	0/5	0/6	0/8	4/19
Tensor rotation	0/5	0/6	0/8	0/19
Modular induction	0/5	0/6	0/8	0/19
Deferred exact checks	0/5	0/6	1/7	0/19
Constructed models	1	3	2	0
Unresolved	4	3	5	19

Table 4: The executed ordered sieve. Criterion cells are newly excluded / remaining. Seconds are recorded pass wall times; 4 or 8 workers per stage are recorded in the ledger. Sphericalization/induction checks retained proof trees for $m \leq 7$ and discovers proofs at $m = 8$. Trees can use nested advanced filters; unfinished retained checks appear in the deferred row. Older residuals are retained. Later dimension-order, character, tensor-rotation and modular discovery targets $m \in \{3, 8\}$. Timeouts remove nothing. This practical order is not claimed optimal.

Retained proof family	Rings
Rank-two fusion subring	631
Formal-codegree d-number obstruction	14753
Zero-spectrum obstruction	4
Complete lower-rank subring descent	144
Involution-module representation obstruction	235
Subcategory dimension divisibility	20
Cyclotomicity	14
Solvability descent and involution-return indicators	4
Noncommuting dimension-field automorphisms	2
Multiplicity-weighted noncommutative induction	7
Commutative signed sphericalization	192
Complete oriented signed sphericalization	376
Direct general-to-positive reduction	70
Further general-to-positive certificates	57
Signed reduction with the common induction test	2
Total	16511

Table 5: Disjoint retained proof families, listed in the order of the corresponding sieve stages. Several signed and positive-dimension families occur within the sphericalization/induction stage. Nested criteria are not counted a second time.

retained columns are unchanged. The stronger framework therefore retains the old exact witnesses rather than treating their special cases as unrelated obstructions.

Every certificate is verified from the original ordered tensor. The checker reconstructs number fields, verifies separating elements and factorizations, recomputes low-column and signed-table possibilities, and checks all integer inequalities. In particular it does not assume that a stored list of branches is exhaustive. For Farkas records it checks (4.12) with Python integers. During certificate verification the floating-point linear-programming entry point is replaced by a function that raises an error if called.

The source revision and tensor hashes fix the counting input. No census is regenerated for the categorification analysis. The positive check reconstructs the listed products, the zesting tensor and scalar coboundary, finite-group character rings, and group-theoretical convolution tensors, including the twisted S_4 cochains and their projective stabilizer factors. Exact Weyl characters reconstruct the G_2 model. Every positive entry, including every multiplicity-one entry, is matched by an explicit unit-fixing basis permutation; the complete 61-record constructor catalogue is checked for omissions. Regression tests compare the native integer routines with straightforward exact implementations, and deliberately corrupted certificates must be rejected.

This is an exact computer-assisted verification, not a formalization in a proof assistant. The categorical input theorems, their hypotheses, and the mathematical completeness argument remain part of the written proof. A checksum proves file integrity, not a mathematical theorem.

Proof of Theorem 1.2. Each negative record is an application of Theorems 5.1 and 3.3, or of Proposition 3.1, Corollary 3.2, or of an exhaustive signed/positive reduction followed by Theorem 4.1 or Theorem 4.3 and the cited rank-four or rank-five tests, with the tensor-sign reduction of Lemma 3.4 and the extension of Theorem 4.5 where recorded. The stronger character-aware system of Theorem 4.7, dimension-order test of Proposition 4.9, and signed refinement of Theorem 4.10 are used where recorded. The exact checker validates all hypotheses and exhausts the required finite alternatives. Thus no excluded tensor has a complex

categorification. Section 6 constructs and tensor-matches all claimed positive examples, with the additional models in Sections 6.4 and 6.6, the higher models of Section 6.8, and complete catalogue in Section 6.9. In particular the spherical existence and pseudo-unitary nonexistence for $R^{(6)}_{2,124}$ use the full-tensor identification and the two cited results of Schopieray. No complement argument is used to assign the positive records. The census and disjoint ledger establish the counts in Table 1. The other records are left undecided, with no assertion of nonexistence or realizability. □

7.2 A realized boundary of the finite relaxations

The ring $R^{(6)}_{2,4}$, realized in Proposition 6.10, has invertible group $\langle g\rangle = C_4$ and two self-dual noninvertibles X, Y. Its rules are

$$\begin{aligned} gX = Y = Xg, \qquad gY = X = Yg, \\ X^2 = Y^2 = 1 + g^2 + 2X + 2Y, \\ XY = YX = g + g^3 + 2X + 2Y. \end{aligned} \tag{7.1}$$

The dimensions are $(1, 1, 1, 1, d, d)$ with $d = 2 + \sqrt{6}$ and $D = 4 + 2d^2 = 24 + 8\sqrt{6}$. This ring is not confused with the Klein-four quadratic ring of the same dimension type: its invertible group is cyclic. Nor is it the zesting example in Proposition 6.2, whose noninvertible square has only one of the two noninvertibles.

Proposition 7.1 (An exact finite feasible branch)**.** *For the ring* (7.1)*, the height-six balanced finite-indicator system has a rational nonnegative feasible point after fixing an integral high-column Gram decomposition. The associated formal list has twenty-two central simples. Thus the necessary system consisting of the chosen low data, Gram equations, algebraic-integrality column bounds, height-six relative traces, duality balance, Galois dimension-count balance, and integer column counts cannot by itself exclude this ring.*

Proof. The file `data/frontier_feasible.json` specifies the complete tensor, low matrix, allowed second indicators, nonnegative integral column counts, and rational values of the trace variables. The checker reconstructs the low-column list and confirms membership, computes every high column, and checks (4.13). It then reconstructs the balanced system and the fixed-count rows and verifies $Ax = b$ and $x \geq 0$ by exact rational arithmetic. The specified column counts sum to sixteen and the low part has six distinct columns, giving twenty-two formal simples. The exact entries, system dimensions, and a successful verification report are included in the supplement. □

Proposition 7.2 (Integral counts in the rotation relaxation)**.** *For $R^{(6)}_{2,4}$, the height-six system of Theorem 4.5 has an exact nonnegative feasible point with integer high-center counts and integer cyclic-character counts. Its system has* 377 *equations in* 893 *variables, with* 60 *nonzero coordinates and formal central rank* 27*.*

Proof. The file `data/frontier_rotation_feasible.json` supplies its low columns, second indicators and exact rational coordinates. The checker independently regenerates all center and rotation rows, verifies nonnegativity and every equality, and checks integrality of all 798 high-center and 55 rotation-count coordinates. Only the inequality slack coordinates are permitted to be rational. Summing the high-center counts and the six low columns gives 27. □

This is a different feasible point from the rank-twenty-two point in Proposition 7.1; they must not be combined into one claimed center. The stronger point shows that adding the

displayed tensor-rotation equations and integrality of these grouped counts does not exclude the cyclic-four tensor, now positively realized in Section 6.6. A separate height-twelve rational search also found no exclusion, but only the height-six exact point is used for the proposition.

Neither proposition constructs a modular datum or a category. In Proposition 7.1 trace counts may be rational. Even the grouped integer counts of Proposition 7.2 specify averaged twist information rather than individual twists satisfying Verlinde and balancing identities. These points are now positive controls for the realized tensor $R_{2,4}^{(6)}$, not grounds for pursuing an exclusion of it. For unresolved tensors, additional information such as individual twist sectors, center fusion coefficients and multiplication compatibility can refine the necessary systems. The distinct point in Section 4.4.4 concerns $R_{3,215}^{(6)}$ and is verified separately. No feasible system is promoted to categorical existence.

The higher-multiplicity residuals include both finite feasible relaxations and bounded signed searches that did not finish. Their diagnostics and complete tensors are supplied separately. The verified feasible points for $R_{2,4}^{(6)}$ must not be silently assigned to other rings. Current quadratic-category work such as [28] is relevant context, but its Klein-four classification is distinct from the cyclic-four conformal-embedding construction above.

7.3 Reproduction and counting data

The top-level command

```
python verify.py -jobs 2
```

checks the frozen input correspondence, positive models, and allocated negative certificates. The command with `python -O` checks that correctness does not depend on Python assertions. Regression controls exercise exact arithmetic, corrupted certificates and the frontier witnesses. Precise dependencies, commands, report scope and search-stop behavior are given in the accompanying verification guide. These are classification checks, not a new enumeration of fusion rings.

The `oeis/` directory contains an eight-term b-file, a CSV, and proposed definition and provenance text. The definition is the number of based-isomorphism classes of rank-six fusion rings of *exact* multiplicity m, without assuming commutativity, self-duality, integrality or categorifiability. The terms are copied from the pinned census. No accession number is assumed and no online submission is claimed. Counting farther and classifying farther are separate achievements.

7.4 Research coverage and the next unresolved stratum

The residual list is derived from the complete decision ledger, not from a hand-selected sample. Multiplicity two is completely decided. The smallest unresolved multiplicity and the exact residual tensors are reported in Section 1; the first such tensor is the priority for further theoretical and automated work. The stage table also shows whether difficulties reappear at larger multiplicities. A criterion applied successfully at one bound remains the same mathematical criterion at the next bound: additional computation alone is not presented as a new theorem.

The module-Gram refinements and relative-center constructions are necessary tests or model constructions with explicit hypotheses. A failed eligibility check does not exclude a ring. An inconclusive Gram factor does not supply a model. For the relative Fibonacci center of TT_3, the categorical construction and exact tensor reconstruction supply the positive conclusion; feasible Gram data by themselves do not.

Literature comparisons distinguish exact fusion-rule matches from shared dimensions. Unitary-only classifications do not exclude arbitrary complex fusion categories, and candidate

modular data are not categorifications unless the source supplies a construction. The same principles govern new models and new necessary criteria: a model must have a justified categorical source and an exact full-tensor match; an obstruction must state its hypotheses and exhaust all branches needed for the claimed categorical structure.

The first residual $R^{(6)}_{3,215}$ has been subjected to the new joint low-product Gram condition of Section 4.6. The program rechecks the 230-equation induction witness, computes all cubic and fourth-order moments and constructs an integral product matrix satisfying the Gram and restriction identities. Its formal rank-22 point survives. This supplies additional exact information about the frontier but no general exclusion or category. The two-sided module Gram identities in Section 5.3 provide a separate refinement with their stated module hypotheses; they are not a license to assume a module category exists.

The Dong–Schopieray bound in Section 3.5 does give a negative result for the first residual, specifically for braided spherical categorification. It leaves the five unbraided existence fields unchanged. The Ng–Rowell–Wen comparison screens every one of the 16612 source tensors against the rank-six ancillary data and performs exact based-ring comparisons after invariant filtering. It finds no new model; the source's largest fusion coefficient is two. Neither failure to match that modular list nor failure of the premodular bound is promoted to nonexistence of an arbitrary fusion category.

These computations are reproduced by

```
python research/m8_frontier/low_product_gram.py --check
python research/m8_frontier/premodular_bound.py
python research/m8_frontier/nrw_rank6_scan.py
```

The programs retain exact certificates and scoped reports alongside their source. They apply their conditions to tensor data rather than using a desired classification outcome as input.

The remaining 47 tensors are unresolved by this release. No literature-wide open-problem claim is attached to that word. In particular, survival of all implemented tests means only that these tests have not decided the input. The paper makes no prediction of counts or categorifiability beyond multiplicity eight.

AI assistance. GPT-6 Astra assisted with research, computation, and manuscript preparation.

References

[1] D. Aasen, E. Lake and K. Walker, *Fermion condensation and super pivotal categories*, arXiv:1709.01941v2 (2018).

[2] M. A. Alekseyev, W. Bruns, J. Dong and S. Palcoux, *Classifying integral Grothendieck rings up to rank 5 and beyond*, accepted in *Letters in Mathematical Physics*; arXiv:2507.07023.

[3] I. Angiono, C. Galindo and G. Mora, *Zestings of Hopf algebras*, arXiv:2505.10447v1 (2025).

[4] S. Bigelow, S. Morrison, E. Peters and N. Snyder, *Constructing the extended Haagerup planar algebra*, Acta Math. **209** (2012), 29–82; arXiv:0909.4099.

[5] P. Bruillard, S.-H. Ng, E. C. Rowell and Z. Wang, *Rank-finiteness for modular categories*, J. Amer. Math. Soc. **29** (2016), 857–881; arXiv:1310.7050v3.

[6] S. Burciu and S. Palcoux, *Structure constants, Isaacs property and extended Haagerup fusion categories*, Comm. Algebra **53** (2025), 1438–1452; arXiv:2210.13936v3, especially Section 5.1.

[7] Y. Choi, D.-C. Lu and Z. Sun, *Self-duality under gauging a non-invertible symmetry*, J. High Energy Phys. **2024**, no. 1, article 142; arXiv:2310.19867v2.

[8] R. Coquereaux, R. Rais and E. H. Tahri, *Exceptional quantum subgroups for the rank two Lie algebras B_2 and G_2*, J. Math. Phys. **51** (2010), 092302; arXiv:1001.5416v1.

[9] C. Delaney, C. Galindo, J. Y. Plavnik, E. C. Rowell and Q. Zhang, *Braided zesting and its applications*, Comm. Math. Phys. **386** (2021), 1–55.

[10] J. Dong and A. Schopieray, *Near-integral fusion*, arXiv:2407.15955v1 (2024).

[11] *Rank-five Grothendieck rings through multiplicity twenty*, manuscript and exact-certificate supplement (2026). The finite-indicator, quadratic-parity and inert-prime parity theorems and Theorem 1.1 refer to this manuscript. Its complete general classification through multiplicity twenty is used only within that proved range.

[12] *Rank-four Grothendieck rings through multiplicity 128*, finite-classification manuscript and exact-certificate supplement (2026). Numbered results refer to the finite-classification manuscript.

[13] C. Edie-Michell, *A complete classification of unitary fusion categories tensor generated by an object of dimension the golden ratio*, arXiv:1904.08909 (2019).

[14] P. Etingof, D. Nikshych and V. Ostrik, *On fusion categories*, Ann. of Math. (2) **162** (2005), 581–642; arXiv:math/0203060.

[15] P. Etingof, S. Gelaki, D. Nikshych and V. Ostrik, *Tensor Categories*, Mathematical Surveys and Monographs **205**, American Mathematical Society, 2015.

[16] P. Etingof, D. Nikshych and V. Ostrik, *Weakly group-theoretical and solvable fusion categories*, Adv. Math. **226** (2011), 176–205; arXiv:0809.3031v2.

[17] D. E. Evans and T. Gannon, *The exoticness and realisability of twisted Haagerup–Izumi modular data*, Comm. Math. Phys. **307** (2011), 463–512; arXiv:1006.1326.

[18] C. Galindo, S.-M. Hong and E. C. Rowell, *Generalized and quasi-localizations of braid group representations*, arXiv:1105.5048v1 (2011).

[19] S. Gelaki, D. Naidu and D. Nikshych, *Centers of graded fusion categories*, Algebra Number Theory **3** (2009), 959–990; arXiv:0905.3117, especially Remark 2.4.

[20] S. Gelaki and D. Naidu, *Some properties of group-theoretical categories*, J. Algebra **322** (2009), 2631–2641; arXiv:0709.4326.

[21] P. Grossman and N. Snyder, *Quantum subgroups of the Haagerup fusion categories*, Comm. Math. Phys. **311** (2012), 617–643; arXiv:1102.2631v2.

[22] P. Grossman and N. Snyder, *The Brauer–Picard group of the Asaeda–Haagerup fusion categories*, Trans. Amer. Math. Soc. **368** (2016), 2289–2331; arXiv:1202.4396v2, Table 1.

[23] M. Izumi, *A Cuntz algebra approach to the classification of near-group categories*, arXiv:1512.04288v2 (2015), especially Example 13.8 and Section 13.3.

[24] M. Izumi, S. Morrison and D. Penneys, *Fusion categories between $\mathcal{C}\boxtimes\mathcal{D}$ and $\mathcal{C}*\mathcal{D}$*, arXiv:1308.5723v1 (2013).

[25] M. Izumi, V. F. R. Jones, S. Morrison and N. Snyder, *Subfactors of index less than 5, part 3: quadruple points*, Comm. Math. Phys. **316** (2012), 531–554; arXiv:1109.3190.

[26] A. Kirillov Jr. and V. Ostrik, *On a q-analogue of the McKay correspondence and the ADE classification of $\widehat{\mathfrak{sl}}_2$ conformal field theories*, Adv. Math. **171** (2002), 183–227; arXiv:math/0101219v3.

[27] Z. Liu, S. Palcoux and Y. Ren, *Classification of Grothendieck rings of complex fusion categories of multiplicity one up to rank six*, Lett. Math. Phys. **112** (2022), article 54; arXiv:2010.10264v5.

[28] Y. Meng and Z. Yu, *Classification of some $\mathbb{Z}/2\mathbb{Z}\times\mathbb{Z}/2\mathbb{Z}$-quadratic fusion categories of rank 6*, arXiv:2608.06064v1, 6 August 2026.

[29] S.-H. Ng, E. C. Rowell and X.-G. Wen, *Classification of modular data up to rank 12*, arXiv:2308.09670v2 (2025), with rank-six ancillary fusion tensors.

[30] S.-H. Ng and P. Schauenburg, *Frobenius–Schur indicators and exponents of spherical categories*, Adv. Math. **211** (2007), 34–71; arXiv:math/0601012.

[31] S.-H. Ng and P. Schauenburg, *Higher Frobenius–Schur indicators for pivotal categories*, Contemp. Math. **441** (2007), 63–90; arXiv:math/0503167v2.

[32] S.-H. Ng, E. C. Rowell, Z. Wang and X.-G. Wen, *Reconstruction of modular data from $SL_2(\mathbb{Z})$ representations*, Comm. Math. Phys. **402** (2023), 2465–2545; arXiv:2203.14829.

[33] S.-H. Ng, E. C. Rowell and X.-G. Wen, *Recovering R-symbols from modular data*, arXiv:2408.02748v1 (2024).

[34] V. Ostrik, *Fusion categories of rank 2*, Math. Res. Lett. **10** (2003), 177–183; arXiv:math/0203255.

[35] V. Ostrik, *On formal codegrees of fusion categories*, Math. Res. Lett. **16** (2009), 895–901; arXiv:0810.3242.

[36] V. Ostrik, *Pivotal fusion categories of rank 3*, Mosc. Math. J. **15** (2015), 373–396; arXiv:1309.4822.

[37] S. Palcoux, *Fusion Ring Census*, public data and verification repository, `https://github.com/sebastienpalcoux/fusion-ring-census`, revision `b86eb6dc6eeeafe5fc5f6906d7b074e0322e9726`, 23 September 2026. Rank-six tensors, counts and provenance are pinned to this revision.

[38] P. Schauenburg, *A higher Frobenius–Schur indicator formula for group-theoretical fusion categories*, arXiv:1502.02906v1 (2015).

[39] A. Schopieray, *Categorification of integral group rings extended by one dimension*, J. London Math. Soc. **108** (2023), 1617–1641, `https://doi.org/10.1112/jlms.12789`. Lemma 3.1 is cited in its published form.

[40] A. Schopieray, *Lie theory for fusion categories: a research primer*, arXiv:1810.09055 (2018).

[41] A. Schopieray, *Non-pseudounitary fusion*, arXiv:2010.02958v1 (2020).

[42] G. Vercleyen and J. K. Slingerland, *On low rank fusion rings*, J. Math. Phys. **64** (2023), 091703; arXiv:2205.15637v4, with ancillary multiplication tables.